

Appendix D:
WORKED EXAMPLES

This Appendix contains several worked examples to illustrate the use of the various assessment methodologies.

Example	Description
D1	Gas compression system: main line qualitative assessment
D2	Gas compression system: main line quantitative assessment
D3	Separation system: main line qualitative assessment
D4	Separation system: main line quantitative assessment
D5	Type 1 SBC assessment
D6	Type 2 SBC assessment
D7	Type 3 SBC assessment
D8	Type 4 SBC assessment

D.1 **EXAMPLE D1: GAS COMPRESSION SYSTEM: QUALITATIVE ASSESSMENT**

This example is based on the assessment of a new design for a gas compression system. The Process Flow Diagram is shown in **Figure D-1**, with the relevant stream data shown in **Table D-1**. The Piping and Instrumentation Diagram is shown in **Figure D-2**.

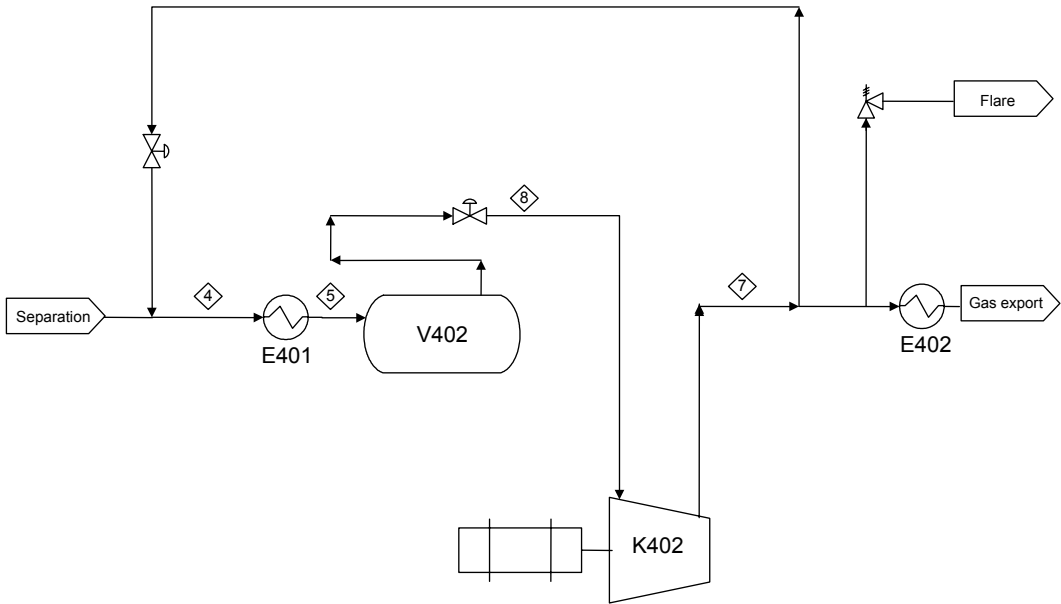


Figure D-1: Example D1: Process Flow Diagram

Stream	4	5	6	7
Vapour Fraction	1	1	1	1
Temperature deg C	141.9	30	30	136.7
Pressure Bar g	25	23.5	23.5	87
Density kg/m³	18	26	23	62
Viscosity cP	0.02	0.01	0.01	0.02
Flow BPD/MMSCFD	51.24	59348	49.29	49.29
Mass flow kg/hr	59358	59358	53482	53482
Mass heat capacity kj/kg-degC	2.49	2.25	2.23	2.74
Molecular weight	23.22	23.22	21.75	21.75
Compressibility	0.96	0.96	0.91	0.91
C _p /C _v	1.22	1.3	1.34	1.33
Heat of vaporisation kj/kg	162	163	151	92

Table D-1: Example D1: Stream data

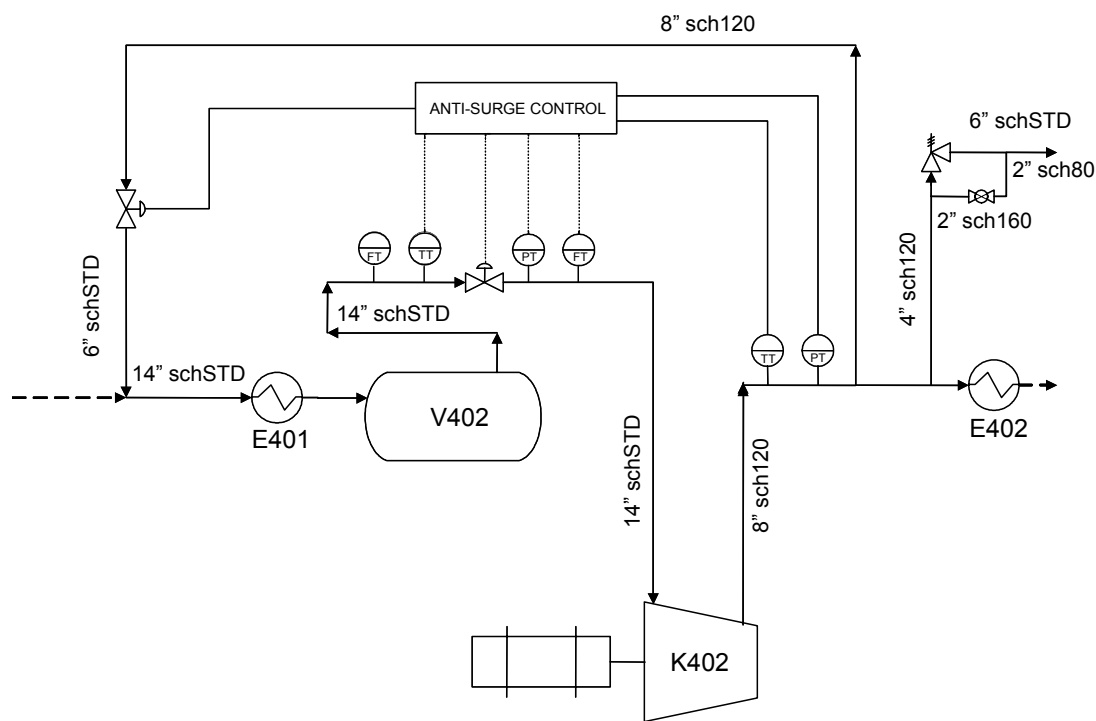
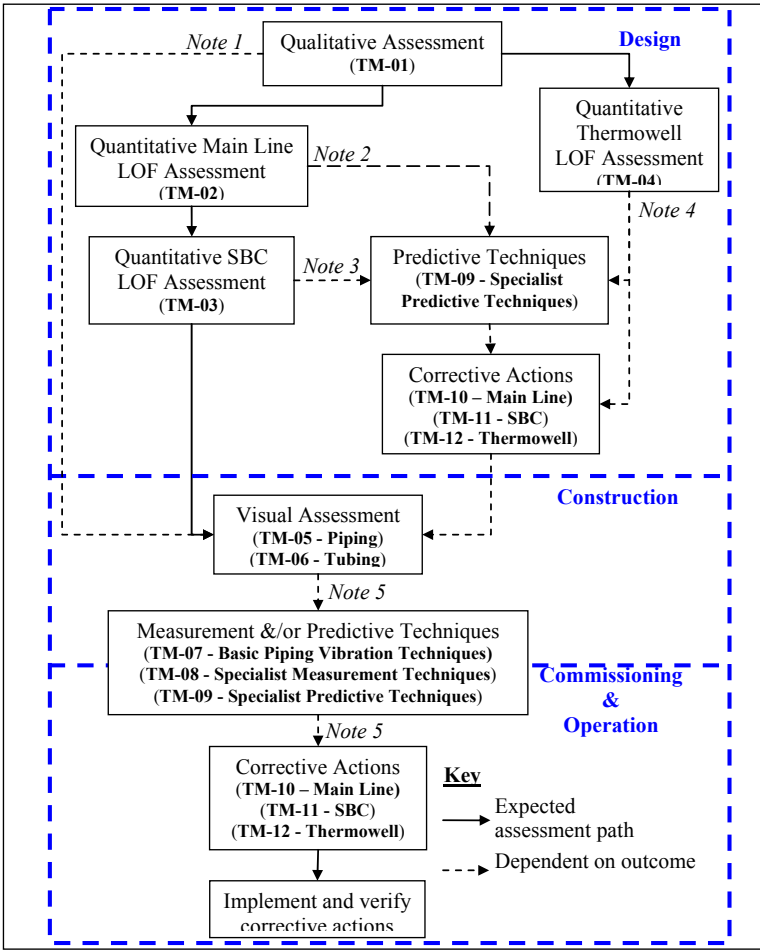


Figure D-2: Example D1: Piping & Instrumentation Diagram

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For the case of a new design, the approach given in **Flowchart 3-1** should be followed.



The first step is to undertake a qualitative assessment as described in **Technical Module TM-01**. This should be undertaken with process and/or operations engineers to ensure that all relevant operational cases are identified and taken into account in the assessment.

The qualitative assessment is undertaken by answering each of the questions in **Table T1-1** in turn, considering the various operational scenarios that may occur.

Item 1: Kinetic energy

Item	Aspect	Applicable process fluid(s)	Likelihood Classification			Potential excitation mechanism(s)
			Low	Medium	High	
1	Is there a high level of kinetic energy (ρv^2) of the process fluid?	All	$\rho v^2 < 5,000 \text{ kg/m s}^2$	between $5,000 = \rho v^2 < 20,000 \text{ kg/m s}^2$	$\rho v^2 > 20,000 \text{ kg/m s}^2$	Flow induced turbulence (All fluids) refer to Section T2.2 Flow induced pulsation (Gases only) refer to Section T2.6

The kinetic energy (ρv^2) for each process stream is calculated and the maximum value obtained is compared with the limits given. This requires knowledge of the stream data (mass flow rate and fluid density) and also the main line internal diameter. In this case on the suction side of K402 (streams 4, 5, 6) the pipework is

14” schedule STD, whilst on the discharge side (stream 7) the pipework is 8” schedule 120.

Stream	4	5	6	7
Calculated ρv^2 (kg/m.s ²)	1909	1321	1213	5191

In this example the maximum value is 5191 kg/m.s², which, when compared to the limits, results in a ‘Medium’ classification.

Note: in some situations the highest value of ρv^2 may not be associated with any of the streams given in a Process Flow Diagram. For example, flow through a recycle, bypass or relief line, whilst not considered in the PFD, may give rise to high levels of process fluid kinetic energy. If there is any doubt (and particularly if none of the process streams given on the PFD have a value greater than 5000 kg/m.s²), then a check should be made on those systems which operate intermittently.

In this case, both flow induced turbulence and flow induced pulsation should be considered.

Item 2: Choked flow / sonic velocity

Item	Aspect	Applicable process fluid(s)	Likelihood Classification			Potential excitation mechanism(s)
			Low	Medium	High	
2	Is choked flow possible or are sonic flow velocities likely to be encountered?	Gas	No		Yes	High frequency acoustic excitation refer to Section T2.7

In this case choked flow is possible under two scenarios: either when (i) the recycle valve is just open or (ii) when the relief valve lifts. This results in a ‘High’ classification. High frequency acoustic excitation must therefore be considered.

Item 3: Machinery

Item	Aspect	Applicable process fluid(s)	Likelihood Classification			Potential excitation mechanism(s)
			Low	Medium	High	
3	Is there any rotating or reciprocating machinery?	All	No	rotating equipment only	reciprocating equipment	Mechanical excitation refer to Section T2.3

The only rotating machinery is the electric motor driven centrifugal compressor K402. This results in a ‘Medium’ classification. Mechanical excitation must therefore be considered.

Item 4: Positive displacement pumps / compressors

Item	Aspect	Applicable process fluid(s)	Likelihood Classification			Potential excitation mechanism(s)
			Low	Medium	High	
4	Are there any positive displacement pumps or compressors?	All	No	Screw/gear type positive displacement machine	reciprocating type positive displacement machine	Pulsation - reciprocating refer to Section T2.4

There are no reciprocating or positive displacement pumps or compressors. This results in a ‘Low’ classification.

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Item 5: Rotating stall

Item	Aspect	Applicable process fluid(s)	Likelihood Classification			Potential excitation mechanism(s)
			Low	Medium	High	
5	Are there any centrifugal compressors which have the potential to operate under rotating stall conditions?	Gas	No	Compressor has stall characteristics but operational restraints in place to ensure that rotating stall is not encountered	Stall rotating condition unknown. Compressor has rotating stall characteristics and may operate at conditions that will give rise to stall conditions	Pulsation - rotating stall refer to Section T2.5

In this example the compressor is known not to exhibit a rotating stall characteristic.

Item 6: Flashing / cavitation

Item	Aspect	Applicable process fluid(s)	Likelihood Classification			Potential excitation mechanism(s)
			Low	Medium	High	
6	Are there any systems which may exhibit flashing or cavitation	Liquid / Multiphase	No		Yes	Cavitation and Flashing refer to Section T2.9

As this is a gas system this excitation mechanism does not apply.

Item 7: Fast acting valves

Item	Aspect	Applicable process fluid(s)	Likelihood Classification			Potential excitation mechanism(s)
			Low	Medium	High	
7	Are there any systems with fast acting opening or closing valves?	All	No		Yes	Surge/ Momentum changes (refer to Section T2.8)

There is only one fast acting opening valve on the system which is the relief valve. This results in a 'High' classification.

Item 8: Intrusive elements

Item	Aspect	Applicable process fluid(s)	Likelihood Classification			Potential excitation mechanism(s)
			Low	Medium	High	
8	Are there intrusive elements in the process stream?	All	No		Yes	Vortex shedding from intrusive elements to refer TM-04

There is one thermowell on the system. Therefore the resulting classification is 'High'.

Item 9: Slug flow

Item	Aspect	Applicable process fluid(s)	Likelihood Classification			Potential excitation mechanism(s)
			Low	Medium	High	
9	Is there a possibility of slug flow?	Multiphase	No		Yes	Slug flow - seek specialist advice

As this is a gas system this excitation mechanism does not apply.

Item 10: History of pipework vibration

Item	Aspect	Applicable process fluid(s)	Likelihood Classification			Potential excitation mechanism(s)
			Low	Medium	High	
10	Is there a history of pipework vibration issues, or are there any systems which are similar to those on another plant which have a known history of pipework vibration issues?	All	No	Yes: however, suitable corrective action in place and validated for the complete operating envelope.	Yes	Known vibration refer to Chapter 4

As this is a new design there is no history of issues. Providing a check is made and there is also no record of piping vibration problems on a similar (operational) plant then a ‘Low’ classification can be made.

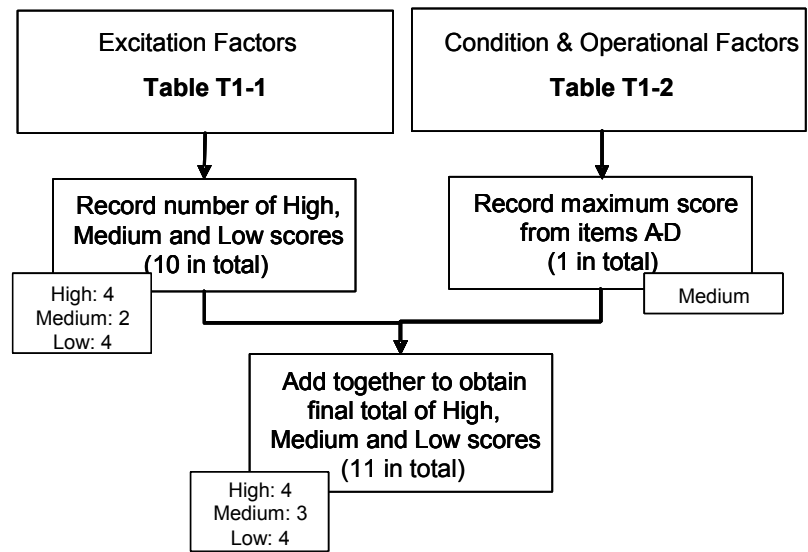
Items A-D: Condition and Operational Factors

Item	Aspect	Applicable process fluid(s)	Likelihood Classification			Contributory factor
			Low	Medium	High	
A	What is the quality of construction?	All	Better than industry standards	At Industry standard	Below industry standards	Build quality
B	What is the effectiveness of the plant maintenance programme (including corrosion management)?	All	Better than industry standards	At industry standard	Below industry standards	Corrosion/ maintenance management
C	Are there any cyclical operations (e.g. batch operation)?	All	No		Yes	Cyclical loading
D	What is the number of unplanned process interruptions in an average year? (this is intended for normal continuous process operations)	All	0-1	2-8	9 or more	Process upsets

As this is a new design items A and B have been assessed as being ‘at industry standard’. There is no cyclical operations and a low number of unplanned process interruptions.

Combination of factors

Flowchart T1-1 is used to combine the various factors and to provide a final ‘score’ for this particular system.



This provides the score for the one system under consideration. If several separate systems had been assessed then each would be individually scored; comparison of the individual system scores would then provide a rank ordering to prioritise the subsequent quantitative assessment.

D.2 EXAMPLE D2: GAS COMPRESSION SYSTEM: QUANTITATIVE ASSESSMENT

This example follows on directly from the qualitative assessment undertaken in **Example D1**. From the results obtained in the qualitative assessment the following excitation mechanisms (based on those scoring Medium or High) should be considered for a quantitative assessment using the relevant methods given in Technical Modules **TM-02** and **TM-04**:

Excitation Mechanism	Technical Module	Section
Flow induced turbulence	TM-02	T2.2
Flow induced pulsation	TM-02	T2.6
High frequency acoustic excitation	TM-02	T2.7
Mechanical excitation	TM-02	T2.3
Surge / momentum changes	TM-02	T2.8
Vortex shedding from intrusive elements	TM-04	

Each of these will be addressed in turn.

D.2.1 Flow Induced Turbulence (see T2.2)

Step 1: Determine ρv^2 (see T2.2.3.1)

For single phase flow $\rho v^2 = (\text{actual density}) \times (\text{actual velocity})^2$

The stream data give the mass flow and density data for the normal full flow condition (stream numbers 4-7). The values of ρv^2 have already been calculated for the qualitative assessment for these stream numbers and are summarised below:

Stream	4	5	6	7
Calculated ρv^2 (kg/m.s ²)	1909	1321	1213	5191

However, there are two further operational cases that need to be considered: (i) recycle operation and (ii) relief conditions. The data for these cases are not usually given in the overall stream data, and must therefore be obtained from other sources.

- (i) Recycle: in the absence of specific information as to the maximum mass flow rate that could be achieved through the recycle line, the maximum compressor discharge flow rate should be used. This is likely to result in a conservative assessment which can then be modified if specific data become available. There are two line sizes to consider:
 - On the compressor discharge side of the recycle valve the recycle line is 8" schedule 120 which gives an internal pipe diameter of 182.4mm. Assuming that the recycle line experiences a maximum flow of 53482 kg/hr with a density of 62 kg/m³ then this would give a value of ρv^2 of 5191 kg/m.s².
 - On the compressor suction side of the recycle valve the recycle line is 6" schedule STD which gives an internal pipe diameter of 154.1mm. Taking a conservative approach and assuming that the gas density is the same as stream 4 (18 kg/m³) with a maximum flow of 53482 kg/hr then this would give a value of ρv^2 of 35294 kg/m.s².
- (ii) Relief: an extract from the valve data sheet is shown below, and gives a flowrate of 49.29 MMscfd once the valve opens, which equates to a mass flow rate of 53482 kg/hr.

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GENERAL	1	Tag Number	PSV -2223		
	2	Service	MP CMFR DSOH		
	3	PID No.	XX-4530-00		
	4	Line Number	Equipment Number	*	XX-1110
	5	Area Classification	IEC Zone 2, Group IIA, T3		
	6	Pipe Size / Pipe Specifications	EN X BN		
	7	Compliance to PED Directive 97 / 23 / EC	Required		
	8				
PROCESS CONDITIONS	9	Fluid	State	HC Vapour	Gas
	10	Required Capacity		49.29	SMt ³ /d
	11	Molecular Weight		21.75	
	12	Oper. Pressure	Set Pressure	87	bar-g
	13	Oper. Temperature	Relieving Temperature	137	°C
	14		Constant	0	bar-g
	15	Back Pressure	Variable	10	bar-g
	16		Total	11	bar-g
	17	% Allowable Overpressure		10	
	18	Overpressure Factor			
	19	Compressibility Factor		0.910	
	20	Latent Heat of Vaporization			kJ/kg
	21	Ratio of Specific Heats		1.33	
	22	Relief Viscosity			cP
	23	Barometric Pressure			
	24	Built Up Back Pressure		1.0 Bar-g	

Note, that this flow rate is the required capacity. The actual installed capacity may be higher when the actual valve is selected. A check should be made when these data become available.

As with the recycle line there are two line sizes / process conditions to consider:

- Upstream of the PSV the relief line is 4" schedule 120 which gives an internal pipe diameter of 92.1mm. The relief line experiences a maximum flow of 53482 kg/hr with a fluid density of 62 kg/m³. This would give a value of ρv^2 of 80203 kg/m.s².
- Downstream of the PSV the line is 6" schedule STD which gives an internal pipe diameter of 154.1mm. The gas density (obtained from a relief system process simulation model) is 4.0 kg/m³ which, with a maximum flow of 53482 kg/hr, gives a value of ρv^2 of 158823 kg/m.s².

A summary of the various ρv^2 values is given below.

Stream	4	5	6	7	Recycle line (compressor discharge)	Recycle line (compressor suction)	Relief line (upstream of PSV)	Relief line (downstream of PSV)
Calculated ρv^2 (kg/m.s ²)	1909	1321	1213	5191	5191	35294	80203	158823

Step 2: Determine Fluid Viscosity Factor (see T2.2.3.2)

As the fluid in this example is gas the fluid viscosity factor (*FVF*) must be calculated; this requires the gas dynamic viscosity (μ_{gas}). This can either be determined from **Figure B-1**, or in this case from the available stream data. Where data are not available (i.e. the recycle and relief lines) then values have been assumed.

Note: that the units required are in Pa.s, whilst often (as in this case) the units for dynamic viscosity are given as cP. To convert from cP to Pa.s multiply by 10⁻³.

The *FVF* factor is then calculated for each case using

$$Fluid\ Viscosity\ Factor = \frac{\sqrt{\mu_{gas}}}{\sqrt{1 \times 10^{-3}}}$$

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Stream	4	5	6	7	Recycle line (compressor discharge)	Recycle line (compressor suction)	Relief line (upstream of PSV)	Relief line (downstream of PSV)
Dynamic viscosity (cP)	0.02	0.01	0.01	0.02	0.02	0.02	0.02	0.01
FVF	0.141	0.1	0.1	0.141	0.141	0.141	0.141	0.1

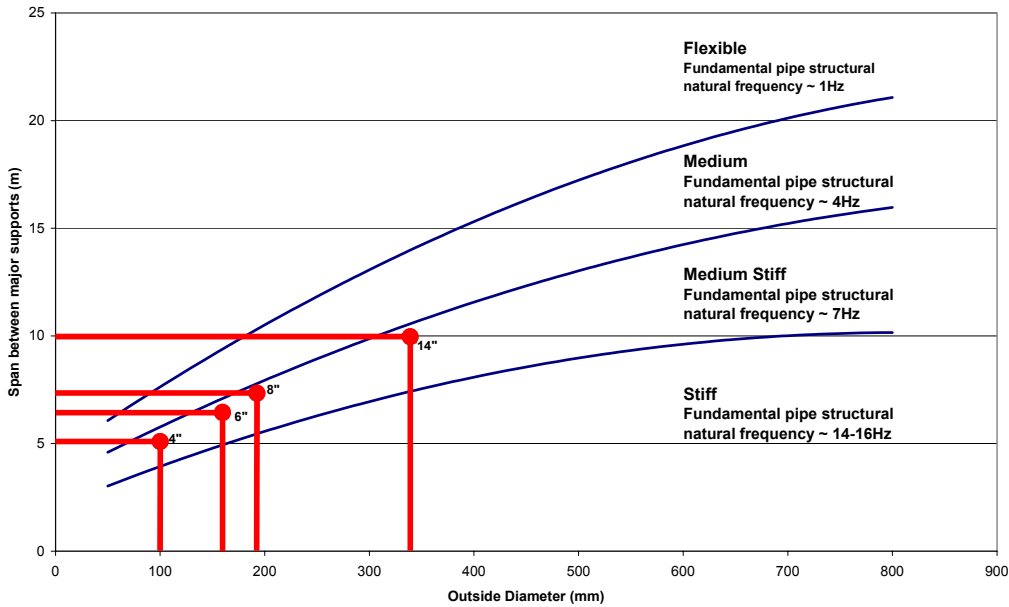
Step 3: Determine Support Arrangement (see T2.2.3.3)

The pipe support arrangement must now be determined. This requires the maximum span lengths between supports to be identified (see guidance in **Appendix B**) and compared with the criteria given in **Table T2-1**. This can be done by (i) working through system isometrics, (ii) walking the lines (on an existing system), or (iii) basing the maximum span length on industry guidance or a particular piping standard or code.

In this example the maximum span lengths have been taken from the project piping standard and are shown below.

Nominal diameter (m)	4"	6"	8"	14"
Maximum span (m)	5.2	6.4	7.3	9.9

These values are then compared with the criteria given in **Table T2-1** (shown graphically below).



In all cases the support classification is “Medium Stiff” (7 Hz).

Step 4: Determine Flow Induced Vibration Factor F_v (see T2.2.3.4)

F_v is determined from the expressions given in **Table T2-2** for the relevant pipe outside diameter and support arrangement. The results are summarised below.

Pipe diameter (schedule)	4" (sch 120)	6" (sch STD)	8" (sch 120)	14" (sch STD)
α	326212	346183	364978	415493
β	-0.9769	-0.9341	-0.9049	-0.8514
F _v	33433	18022	38483	19061

Step 5: Calculation of Likelihood of Failure (LOF) (see T2.2.3.5)

Finally the LOF for each line is calculated using:

Flow Induced Turbulence LOF = (ρv² / Fv) FVF

The results are summarised below.

Stream (Sub system) Pipe dimensions	4 (supply to cooler) 14" (sch STD)	5 (supply to suction scrubber) 14" (sch STD)	6 (compressor suction) 14" (sch STD)	7 (compressor discharge) 8" (sch 120)	Recycle line (compressor discharge) 8" (sch 120)	Recycle line (compressor suction) 6" (sch STD)	Relief line (upstream of PSV) 4" (sch 120)	Relief line (downstream of PSV) 6" (sch STD)
ρv² (kg/m.s²)	1909	1321	1213	5191	5191	35294	80203	158823
FVF	0.141	0.1	0.1	0.141	0.141	0.141	0.141	0.1
Fv	19061	19061	19061	38483	38483	18022	33433	18022
LOF	0.014	0.007	0.006	0.019	0.019	0.276	0.338	0.881

D.2.2 Flow Induced Pulsation (see T2.6)

The assessment procedure is shown in Flowchart T2-4.

Step 1: Determine critical side branch diameter

d_crit = 1000 ((400 / (π ρ v²)))^0.5

This requires the values of ρv² calculated previously.

Stream (Sub system) Pipe dimensions	4 (supply to cooler) 14" (sch STD)	5 (supply to suction scrubber) 14" (sch STD)	6 (compressor suction) 14" (sch STD)	7 (compressor discharge) 8" (sch 120)	Recycle line (compressor discharge) 8" (sch 120)	Recycle line (compressor suction) 6" (sch STD)	Relief line (upstream of PSV) 4" (sch 120)	Relief line (downstream of PSV) 6" (sch STD)
Calculated ρv² (kg/m.s²)	1909	1321	1213	5191	5191	35294	80203	158823
d_crit (mm)	258	310	324	157	157	60	40	28

Step 2: Identify side branches on each main line with ID ≥ d_crit

Once the critical side branch diameter is calculated then any side branches with an internal diameter greater or equal to d_crit must be identified.

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- Supply to cooler: the only side branch is the recycle line (which would act as a deadleg if the recycle valve is shut). However, the internal diameter of the recycle line is 154mm (6" sch STD) which is less than d_{crit} .
- Supply to suction scrubber: no side branches exist with a d_{crit} greater than 310mm.
- Compressor suction: although there are connections for the flow and pressure instruments these are all 2" nominal bore or less. There are no side branches with a d_{crit} greater than 324mm.
- Compressor discharge: the d_{crit} is 157mm; the recycle line (8" sch 120) has an internal diameter of 183mm and is therefore a potential problem. The internal diameter of the relief line is 92.1mm which is below the d_{crit} threshold.
- Under recycle conditions there is flow through the recycle line, and therefore any side branches off the recycle line need to be identified. However, in this case, there are none.
- Under relief conditions there is flow through the relief line. The deadleg side branch caused by the 2" bypass around the PSV with the 2" valve locked closed has an internal diameter of 43mm (upstream of the PSV) and 49mm (downstream of the PSV). Both of these are greater than the relevant d_{crit} (40mm and 28mm respectively) and are therefore potential issues.

Step 3: Determine Reynolds Number

For the remaining two side branches the Reynolds Number of the flow in the main line is calculated using:

$$Re = \frac{\rho v D_{Char}}{1000 \mu}$$

Where D_{Char} is internal diameter of main line

Side branch	1	2	3
Description	Recycle line (8" sch 120)	2" PSV bypass (2" sch 160)	2" PSV bypass (2" sch 80)
Main line	Compressor discharge (8" sch 120)	Relief line (4" sch 120)	6" (sch STD)
Fluid density (kg/m ³)	62	62.0	4.0
Fluid velocity in main line (m/s)	9.2	36.0	199.3
D_{int} (mm)	183	92.1	154
Dynamic viscosity (Pa.s)	2e-5	2e-5	1e-5
Re	5.19e6	1.03e7	1.23e7

In all cases the Reynolds Number is below 1.6×10^7 and therefore S_1 needs to be calculated.

Step 4: Calculate Strouhal Number and Excitation Frequency

$$S_1 = 0.420 \left(\frac{d_{int}}{D_{int}} \right)^{0.316} \left(\frac{v}{c} \right)^{-0.083} \left(\frac{Re}{10^6} \right)^{-0.065}$$

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Where c is the speed of sound in the gas, given by (see **Appendix B**):

$$c = \sqrt{\frac{\gamma R T_e}{M_w}}$$

The temperature of the gas drops across the PSV and so the speed of sound must be calculated for both the upstream and downstream cases as follows:

Side branch	1	2	3
Description	Recycle line (8" sch 120)	2" PSV bypass (2" sch 160)	2" PSV bypass (2" sch 80)
Main line	Compressor discharge (8" sch 120)	Relief line (4" sch 120)	6" (sch STD)
Molecular weight M_w	21.75	21.75	21.75
γ (Cp/Cv)	1.33	1.33	1.33
R (J/K.kmol)	8314	8314	8314
T_e (deg C)	136.7	136.7	88.0*
c (m/s)	456.4	456.4	428.4

*from relief system process simulation

Side branch	1	2	3
Description	Recycle line (8" sch 120)	2" PSV bypass (2" sch 160)	2" PSV bypass (2" sch 80)
Main line	Compressor discharge (8" sch 120)	Relief line (4" sch 120)	6" (sch STD)
Side branch internal diameter (d_{int}) (mm)	183	43	49
Main line internal diameter (D_{int}) (mm)	183	92	154
c (m/s)	456.4	456.4	428.4
Fluid velocity in main line (m/s)	9.2	36.0	199.3
Re	5.19e6	1.03e7	1.23e7
S_f	0.522	0.350	0.265
Ratio d_{int}/D_{int}	1.0	0.47	0.32
Strouhal Number, S	1.044	0.350	0.265

Note, that for side branch 1 the S_f value is multiplied by 2 due to the d_{int}/D_{int} ratio.

The next step is to calculate the fundamental Strouhal Number (S) and the fundamental excitation frequency (F_v) for each sidebranch, using:

$$F_v = \frac{Sv}{d_{int}}$$

Sidebranch	1	2	3
Description	Recycle line (8" sch 120)	2" PSV bypass (2" sch 160)	2" PSV bypass (2" sch 80)
Main line	Compressor discharge (8" sch 120)	Relief line (4" sch 120)	6" (sch STD)
Sidebranch internal diameter (d_{int}) (mm)	183	43	49
Main line internal diameter (D_{int}) (mm)	183	92	154
Strouhal Number	1.044	0.350	0.265
F_v (Hz)	52.2	293.4	1076.8

Step 5: Calculate fundamental acoustic natural frequency of side branch

The next step is to calculate the fundamental acoustic natural frequency of the branch using:

$$F_s = 0.206 \frac{c}{L_{branch}}$$

Where L_{branch} is the length of the side branch and

The length of the two side branches are determined from the system isometrics. In each case the length is the total distance between the connection to the main pipe at one end and the closed valve at the other.

Side branch	1	2	3
Description	Recycle line (8" sch 120)	2" PSV bypass (2" sch 160)	2" PSV bypass (2" sch 80)
Main line	Compressor discharge (8" sch 120)	Relief line (4" sch 120)	6" (sch STD)
L_{branch} sidebranch length (m)	5.1	0.3	1.1
c (m/s)	456.4	456.4	428.4
F_s	18.4	313.4	80.2

Step 6: Obtain LOF score

Finally, the ratio of F_v/F_s is calculated:

Side branch	1	2	3
Description	Recycle line (8" sch 120)	2" PSV bypass (2" sch 160)	2" PSV bypass (2" sch 80)
Main line	Compressor discharge (8" sch 120)	Relief line (4" sch 120)	6" (sch STD)
F_v	52.2	293.4	1076.8
F_s	18.4	313.4	80.2
F_v/F_s	2.83	0.94	13.42

Therefore sidebranch 2 scores an LOF of 0.29, while sidebranches 1 and 3 score an LOF of 1.

D.2.3 High Frequency Acoustic Excitation (see T2.7)

There are two cases to consider:

- (i) When the relief valve lifts
- (ii) When the recycle valve opens

The assessment method is shown in **Flowchart T-2-5**. Note that a more comprehensive acoustic fatigue assessment is shown in **Example D-3**.

Step 1: PWL Calculation

The first step is the calculation of the sound power level (PWL) using:

$$PWL \text{ (source)} = 10 \log_{10} \left[\left(\frac{P_1 - P_2}{P_1} \right)^{3.6} W^2 \left(\frac{Te}{Mw} \right)^{1.2} \right] + 126.1 + SFF$$

In both cases sonic flow does not exist and therefore SFF=0.

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The results of the calculation of PWL are given below. For the relief valve the relevant data are taken from the valve data sheet with a worst case assumption made that the downstream pressure is 1 Bar absolute; for the recycle valve the worst case condition is taken which assumes the highest mass flowrate combined with the maximum pressure drop across the valve.

Valve	Relief	Recycle
P_1 (Bar g)	98	87
P_2 (Bar g)	0	25
W (kg/hr)	53482	53482
T_e (deg C)	137	136.7
Mw	21.75	21.75

Converting to appropriate units and calculating the PWL for each valve:

Valve	Relief	Recycle
P_1 (Pa absolute)	9 900,000	8 800,000
P_2 (Pa absolute)	100	2600
W (kg/s)	14.86	14.86
T_e (K)	410	410
Mw	21.75	21.75
PWL (dB)	164.7	159.4

The source sound power levels of both sources is above 155 dB.

Examination of the recycle valve data sheet (below) shows the valve is fitted with a multi-path, multi-stage trim which, according to the valve manufacturer, gives a reduction in external sound pressure level of approximately 30dB (118 dB-88.2dB).

System Noise with Stages/Turns Supplied			
	dBA		
*System Noise	81.3		
System Noise versus Trim Type			
Ported Cage	118		
Trim with 16 Stages/Turns	88.2		
Notes:			
1. Sound Pressure Levels shown has a +0, -5 dBA tolerance band.			
2. System Noise is the noise produced by the fluid exiting the valve trim and the valve outlet			

If this reduction is applied to the PWL then the PWL of the recycle valve falls below 155dB and therefore the main line LOF for the recycle line for high frequency acoustic excitation is set to 0.29 as shown in **Flowchart T2-5**.

Conversely the relief valve has no low noise trim and therefore the PWL remains unaltered at 164.7dB and the methodology given in **Flowchart T2-5** is followed:

Step 2: LOF Calculation

- Go to next welded discontinuity (e.g. SBC, welded tee, welded support)

From inspection of the system drawings the first welded discontinuity downstream of the source is the 2" bypass line. This is a weldolet connection.

- Calculate the PWL in the main line at the discontinuity accounting for attenuation

PWL at the discontinuity is calculated using:

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$$\text{PWL (discontinuity)} = \text{PWL (source)} - 60 \frac{L_{dis}}{D_{int}}$$

In this case the connection is 0.8m downstream of the source, so :

2" SBC connection	Value
L_{dis} (m)	0.8
D_{int} (mm)	154
$60 \times L_{dis} / D_{int}$	0.312
PWL source (dB)	164.7
PWL discontinuity (dB)	164.4

- Are there any additional sources? In this case, no.
- Is PWL > 155dB? In this case, yes. Continue to **Flowchart T2-6**.
- Calculate N :

2" SBC connection	Value
D_{ext} (mm)	168.3
T (mm)	7.11
A	0.93989
S	68.229
B	152.207
$\log_{10} N$	9.9026
N	7.99E9

- Calculate D_{ext}/d_{ext} ($= 168.3 / 60.3$) = 2.791
- As this is <10 calculate FLM_1 ($= 1.2133$)
- Calculate new N ($= 7.99E9 \times 1.2133$) = 9.70E9
- As the connection is a weldolet, calculate $FLM_2 = 0.2009$
- Calculate new N ($= 9.70E9 \times 0.2009$) = 1.95E9
- The piping material is not duplex therefore calculate $L_f = 0.31$
- As $L_f < 0.5$ then LOF = 0.29

The assessment would then return to **Flowchart T2-5** and move to the next discontinuity on the line downstream.

D.2.4 Mechanical Excitation (see T2.3)

The LOF value is dependent on the maximum LOF from **Table T2-3**. The only source is the electric motor driven centrifugal compressor. The compressor would score an LOF of 0.2 while the electric motor (which is > 15kW) scores 0.4. Therefore the overall LOF value to be used is 0.4.

In this case this LOF value would be applied to the suction line (as far as the suction scrubber) and the discharge line (as far as the cooler). Note that as the recycle line is connected to the discharge line before the cooler (and hence potentially subject to vibration transmission from the discharge line) then the recycle line would also score an LOF of 0.4 for mechanical excitation.

D.2.5 Surge / Momentum Changes (see T2.8)

As the system fluid is gas, only one of the mechanisms (dry gas rapid valve opening) applies – see **T2.8.3.1**. There is only one fast opening valve on the system (the relief valve).

Step 1: Peak Force Calculation

For each valve the peak force is calculated using:

$$F_{\max} = \frac{W}{1000} \sqrt{\frac{2 \cdot \gamma \cdot R \cdot T_e}{(\gamma + 1) \cdot M_w}}$$

Valve	Relief valve
W (kg/s)	14.86
γ (Cp/Cv)	1.33
R (J/K.kmol)	8314
T_e (deg K)	410
M_w	21.75
$F_{\max}$ (kN)	6.28

Step 2: Limit Force Calculation

The next step is to calculate the limit force using:

$$F_{\lim} = (16.8 \times \Psi^3 - 1.81 \times \Psi^2 + 525 \times \Psi + 25.3) \times D_{\text{ext}} \times \theta \times \pi \times D_{\text{int}}^2 / (4 \times 10^9) \text{ (kN)}$$

Valve	Relief valve	
	4" sch 120	6" sch STD
T (mm)	11.1	7.11
T sch 40 (mm)	6.02	7.11
Ψ	1.843	1
D_{ext} (mm)	114.3	168.3
D_{int} (mm)	92.1	154.1
θ (medium stiff support)	2	2
$F_{\lim}$ (kN)	1.67	3.55

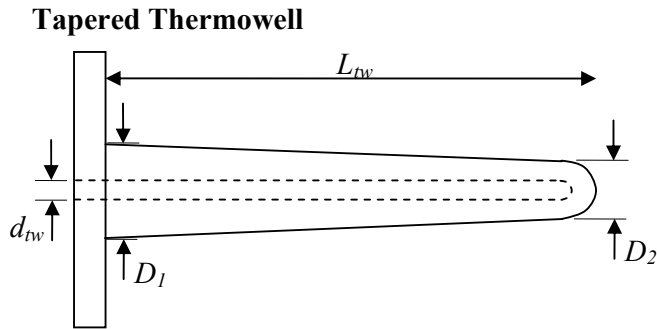
Step 3: LOF Calculation

Finally, the LOF value is calculated using $F_{\max} / F_{\lim}$

Valve	Relief valve	
	4" sch 120	6" sch STD
$F_{\max}$ (kN)	6.28	6.28
$F_{\lim}$ (kN)	1.67	3.55
LOF	3.77	1.77

D.2.6 Vortex Shedding from Intrusive Elements (TM-04)

There is a single thermowell associated with TT-001. Dimensions (taken from the manufacturer's drawings) are shown below:



d_{tw} : 8mm
 D_1 : 26.5mm
 D_2 : 18.0mm
 L_{tw} : 225mm
 E_{tw} : 207E9 N/m²
 ρ : 7850 kg/m³

The main steps in the assessment are as follows:

Step 1: Predict Thermowell Structural Natural Frequency

This is calculated using the following expression for a tapered thermowell:

$$f_n = \frac{1.12 D_1}{1000 L_{tw}^2} \sqrt{\frac{E_{tw}}{\rho} \left(\frac{k^4 + 5k^3 + 15k^2 + 35k + 70 - 126\delta^4}{5353k^2 + 2142k + 513 - 8008\delta^2} \right)} \quad (\text{Hz})$$

Thermowell	Value
d_{tw} (mm)	8
D_1 (mm)	26.5
D_2 (mm)	18
L_{tw} (m)	0.225
E_{tw} (N/m ²)	207E9
ρ (kg/m ³)	7850
K	0.679245
δ	0.301887
f_n (Hz)	497.9

Step 2: Parent Pipework Wall Thickness Modifier

The parent pipe is 14” schedule STD. The pipe wall thickness (9.5mm) is less than schedule 40 (11.1 mm) and therefore a wall thickness modifier (F_M) of 0.42 is selected (assuming the connection does not have 4 way welded gusset plates).

Step 3: Strouhal Number

Firstly, the Reynolds Number is calculated, using

$$\text{Re} = \frac{\rho v D_{Char}}{1000 \mu}$$

where the characteristic dimension D_{Char} is D_2

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Thermowell	Value
Fluid density (kg/m ³)	23
Fluid velocity in main line (m/s)	7.26
D ₂ (m)	0.018
Dynamic viscosity (Pa.s)	1e-5
Re	3.01E5

A Reynolds Number of 3.01E5 gives a conservative value of 0.25 for the Strouhal Number (see **Section T4.2.3**)

Step 4: Vortex Excitation Frequency

The vortex excitation frequency is calculated using:

$$F_v = \frac{1000 \times S \times v}{D_{Char}} \quad (\text{Hz})$$

With D_{Char} taken as D_2 (18mm). This gives $F_v = 100.8$ Hz.

Step 5: LOF Calculation

The value of $F_v/(f_n \times F_m)$ is calculated (= 0.48). This is <0.8 and therefore the LOF is set to 0.29.

D.2.7 Summary and Interpretation of Main Line LOF Scores

Stream (Sub system) Pipe dimensions	4 (supply to cooler) 14" (sch STD)	5 (supply to suction scrubber) 14" (sch STD)	6 (compressor suction) 14" (sch STD)	7 (compressor discharge) 8" (sch 120)	Recycle line (compressor discharge) 8" (sch 120)	Recycle line (compressor suction) 6" (sch STD)	Relief line (upstream of PSV) 4" (sch 120)	Relief line (downstream of PSV) 6" (sch STD)
Flow induced turbulence	0.01	0.01	0.01	0.02	0.02	0.28	0.34	0.88
Flow induced pulsation	0.2	0.2	0.2	1.0 (recycle line)	n/a	n/a	0.29	1.0 (2"bypass)
High frequency acoustic excitation	n/a	n/a	n/a	n/a	0.29	0.29	n/a	0.29
Mechanical excitation	n/a	n/a	0.4	0.4	0.4	n/a	0.4	n/a
Surge / momentum changes	n/a	n/a	0.22	n/a	n/a	n/a	3.77	1.77

Main Line LOF ≥ 1.0

A summary of the actions required for a main line LOF score ≥ 1.0 are given below.

Score	Action	Technical Module
LOF ≥ 1.0	The main line shall be redesigned, resupported or a detailed analysis of the main line shall be conducted, and vibration monitoring of the main line shall be undertaken (Note 1)	TM-09 TM-07/TM-08
	Corrective actions shall be examined and applied as necessary	TM-10
	Small bore connections on the main line shall be assessed.	TM-03
	A visual survey shall be undertaken to check for poor construction and/or geometry and/or support for the main line and/or potential vibration transmission to neighbouring pipework.	TM-05 TM-06

There are several lines that score an LOF ≥ 1.0 :

- The compressor discharge line scores an LOF of 1.0 for flow induced pulsation. This is due to the recycle line acting as a deadleg when the recycle valve is shut. The screening method could be used to see whether varying the length of the recycle line deadleg could reduce the LOF score. Alternatively, a more detailed calculation could now be performed using an acoustic simulation of the recycle line and compressor discharge pipework to accurately predict the acoustic natural frequencies and the excitation frequencies of the recycle line (see **Section T9.5** and reference **[T9-6]** in particular). This would identify whether coincidence will occur for the range of flow rates anticipated, and whether the resulting shaking forces are unacceptable.

Remedial measures should be investigated as outlined in **T10.6**. For example, at the design stage it may be feasible to shorten the length of the recycle line between the discharge line and the recycle valve by re-locating the recycle valve. This would have the effect of increasing the acoustic natural frequencies of the recycle line such that any predicted coincidence no longer occurs. Vibration monitoring of the recycle and discharge line should also be considered during operation.

Finally, all small bore connections on the discharge line and the 'deadleg' recycle line should also be assessed.

- Similarly the relief line downstream of the PSV scores an LOF of 1.0 due to the 2" branch downstream of the PSV acting as a deadleg when the PSV lifts. In this case the screening method could be used to see whether varying the length of the 2" deadleg could reduce the LOF score. Alternatively, as with the recycle line above, a more detailed assessment of the 2" branch pipework could be undertaken to identify the range of coincidence between the fundamental excitation frequency and the acoustic natural frequencies of the 2" deadleg (see **Section T9.5**).

It might be argued that in this case the high flow velocities which give rise to this potential mechanism will only occur for a short time period until the system is depressurised, and therefore the possibility of a fatigue failure may be limited. However, in this case it would be prudent to adopt a conservative approach (particularly for such a safety critical system).

- The relief line upstream and downstream of the relief valve scores an LOF greater than 1.0 due to the forces generated when the relief valve lifts. In this case some form of detailed assessment should be undertaken to ensure that the pipework and associated supports can withstand the associated dynamic loads. Such an analysis is beyond the scope of this document.

Main Line LOF ≥ 0.5 and < 1.0

A summary of the actions required for a main line LOF score ≥ 0.5 are given below.

Score	Action	Technical Module
1.0 > LOF ≥ 0.5	The main line should be redesigned, resupported or a detailed analysis of the main line should be conducted, or vibration monitoring of the main line should be undertaken (Note 1)	TM-09 TM-07/TM-08
	Corrective actions should be examined and applied as necessary	TM-10
	Small bore connections on the main line shall be assessed.	TM-03
	A visual survey shall be undertaken to check for poor construction and/or geometry and/or support for the main line and/or potential vibration transmission to neighbouring pipework.	TM-05 TM-06

There is one main line that scores an LOF ≥ 0.5 :

- The relief line downstream of the PSV has an LOF of 0.88. This is due to the relatively high flow velocity through the line giving rise to a high level of turbulent energy. In this case it may be feasible to increase the stiffness of the piping (at present it is assessed as 'medium stiff' – changing the assessment to 'stiff' would reduce the LOF value). Small bore connections on this line should also be assessed.

Main Line LOF ≥ 0.3 and < 0.5

A summary of the actions required for a main line LOF score ≥ 0.3 are given below.

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Score	Action	Technical Module
0.5 > LOF ≥ 0.3	Small bore connections on the main line should be assessed.	TM-03
	A visual survey should be undertaken to check for poor construction and/or geometry and/or support for the main line and/or potential vibration transmission from other sources.	TM-05
		TM-06

There are several lines that score an LOF ≥ 0.3:

- The relief line upstream of the PSV has an LOF of 0.34 due to flow induced turbulence. No main line issues are anticipated, but a small bore connection assessment should be undertaken.
- Mechanical excitation potentially affects the compressor suction, discharge and the recycle line, and the relief line downstream of the compressor. Again, no main line issues are anticipated, but a small bore connection assessment should be undertaken.

Main Line LOF < 0.3

A summary of the actions required for a main line LOF score < 0.3 are given below.

Score	Action	Technical Module
LOF < 0.3	A visual survey should be undertaken to check for poor construction and/or geometry and/or support for the main line and/or potential vibration transmission from other sources.	TM-05 TM-06

For all main lines a walkdown should be conducted during the construction phase to ensure that the as-built arrangement is fit for purpose, using the guidance given in **TM-06** and **TM-07**.

D.2.8 Summary and Interpretation of Thermowell LOF Score

Stream (Sub system) Pipe dimensions	⁴ (supply to cooler) 14" (sch STD)	⁵ (supply to suction scrubber) 14" (sch STD)	⁶ (compressor suction) 14" (sch STD)	⁷ (compressor discharge) 8" (sch 120)	Recycle line (compressor discharge) 8" (sch 120)	Recycle line (compressor suction) 6" (sch STD)	Relief line (upstream of PSV) 4" (sch 120)	Relief line (downstream of PSV) 6" (sch STD)
Vortex shedding from intrusive elements	n/a	n/a	0.29	n/a	n/a	n/a	n/a	n/a

No issues are anticipated.

D.3 **EXAMPLE D3: SEPARATION SYSTEM: QUALITATIVE ASSESSMENT**

This example is based on the assessment of an existing separation system where a large increase in water production is being considered. The Process Flow Diagram is shown in **Figure D-3**, with the relevant stream data shown in **Table D-2** (for the original case) and **Table D-3** for the revised case. The Piping and Instrumentation Diagram is shown in **Figure D-4**.

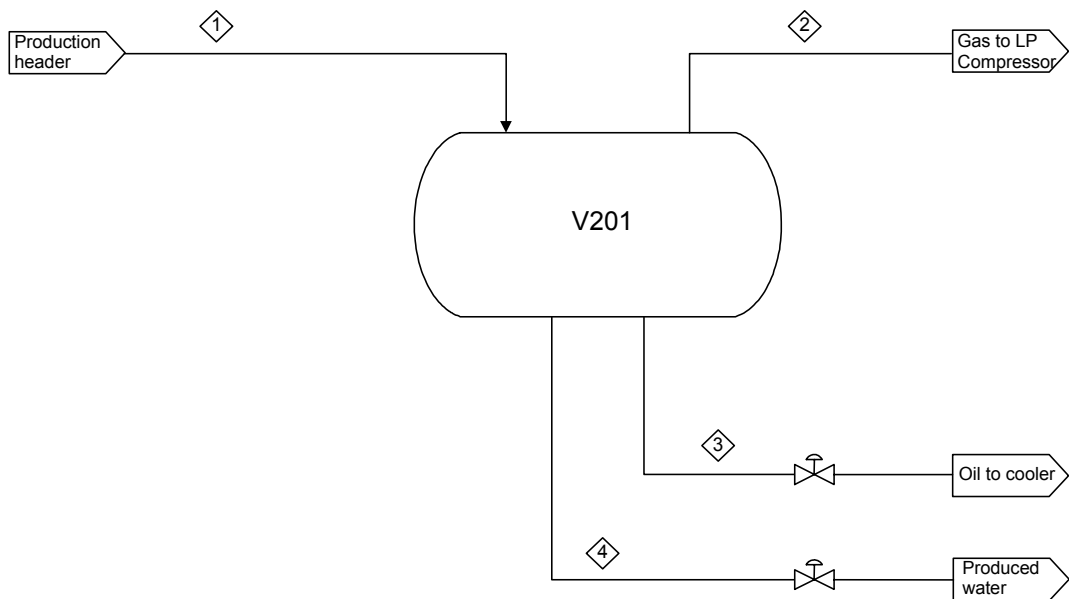


Figure D-3: Example D3: Process Flow Diagram

Stream	1	2	3	4
Temperature deg C	50	50	50	50
Pressure Bar g	5.5	5.5	5.5	5.5
Gas flow m ³ /hr	5436	5436		
Gas density kg/m ³	5	5		
Oil flow m ³ /hr	326		326	
Oil density kg/m ³	930		930	
Water flow m ³ /hr	133			133
Water density kg/m ³	988			988
Viscosity cP		0.01	39.35	0.55
Mass flow kg/hr	461764	27180	303180	131404
Molecular weight	28.09	21.99	38.29	18.05
Compressibility		0.98		
Cp/Cv	1.01	1.24	1.08	1.16

Table D-3: Example D3: Stream data (original)

Stream	1	2	3	4
Temperature deg C	50	50	50	50
Pressure Bar g	5.5	5.5	5.5	5.5
Gas flow m ³ /hr	5436	5436		
Gas density kg/m ³	5	5		
Oil flow m ³ /hr	326		326	
Oil density kg/m ³	930		930	
Water flow m ³ /hr	532			532
Water density kg/m ³	988			988
Viscosity cP		0.01	39.35	0.55
Mass flow kg/hr	855976	27180	303180	525616
Molecular weight	28.09	21.99	38.29	18.05
Compressibility		0.98		
Cp/Cv	1.01	1.24	1.08	1.16

Table D-4: Example D3: Stream data (revised – increased water cut)

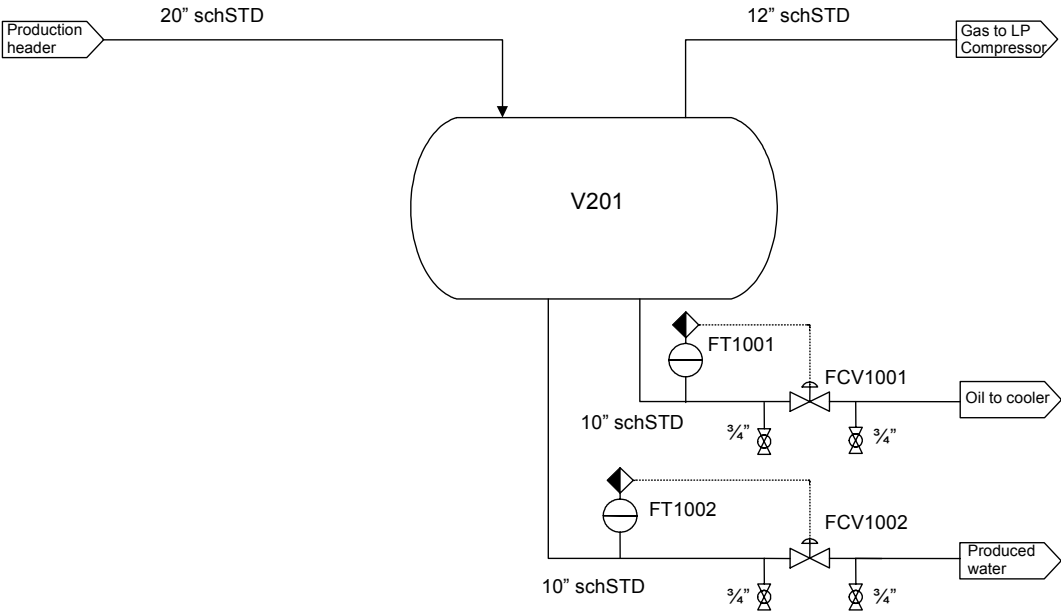
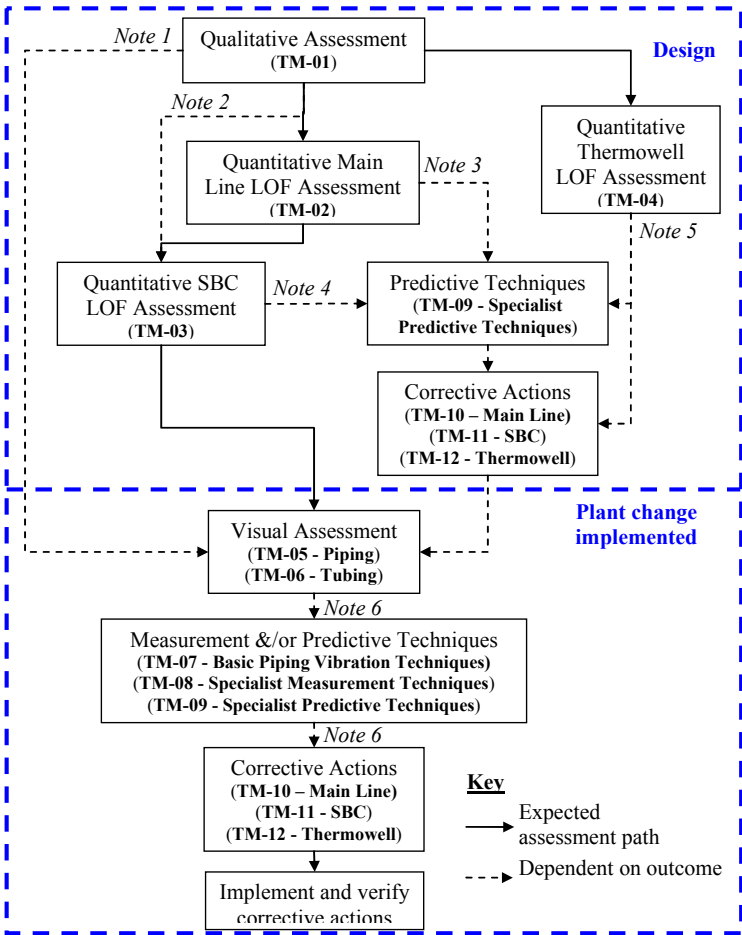


Figure D-4: Example D3: Piping & Instrumentation Diagram

For the case of a change to an existing plant the approach given in **Flowchart 3-3** should be followed.

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The first step is to undertake a qualitative assessment as described in **Technical Module TM-01**. This should be undertaken with process and/or operations engineers to ensure that all relevant operational cases are identified and taken into account in the assessment.

Note: For this example it is assumed that the existing pipework and process conditions have already been assessed for vibration induced fatigue, and that any existing vibration issues have been addressed, with suitable mitigation measures in place.

The qualitative assessment is undertaken by answering each of the questions in **Table T1-5** in turn, considering the various operational scenarios that may occur.

Item 1: Increase in flow velocities and/or fluid densities

Item	Description	If Yes - Potential Issues
1	Will the modification result in one or more of the following •An increase in flow velocities by more than 5% over previous operational experience? •An increase in fluid density by more than 10% over previous operational experience?	•Flow induced turbulence (all fluids), refer to Section T2.2 •Flow induced pulsation (gases systems only), refer to Section T2.6 •Vortex shedding from intrusive elements (all fluids), refer to TM-04 •Surge/Momentum Change refer to Section T2.8

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In this example the only change is the increase in the water mass flow, which affects streams 1 and 4. There are no changes to fluid densities. The increase in fluid velocity is proportional to the increase in volumetric flow rate.

Stream	1	4
Volumetric flow rate (original case) m ³ /hr	5895 (summation of oil, gas and water)	133
Volumetric flow rate (revised case) m ³ /hr	6294 (summation of oil, gas and water)	532
% increase	6.8	300.0

In this example both streams will experience an increase in fluid velocity of over 5%. There are no intrusive elements in the system and therefore only the following potential issues need to be considered for these two process streams:

- Flow induced turbulence
- Surge/momentum changes

Item 2: Change in gas properties

Item	Description	If Yes - Potential Issues
2	For a gas system, will the modification result in one or more of the following: •A change in the molecular weight of the gas by more than ± 5% from previous maximum/minimum operational experience? •A change to the temperature of the gas by more than ± 5% from previous maximum/minimum operational experience? •A change to the ratio of specific heats (C_p/C_v) of the gas by more than ± 5% from previous maximum/minimum operational experience?	For all systems: •Pulsation - Flow induced excitation, refer to Section T2.6 If there is a centrifugal compressor: •Pulsation - rotating stall (gas systems only) refer to Section T2.5 If there is a reciprocating compressor: •Pulsation – reciprocating compressor (gas systems only) refer to Section T2.4

No changes are made to the gas properties and therefore no potential issues are identified.

Item 3: Change in liquid properties

Item	Description	If Yes - Potential Issues
3	For a liquid system incorporating a reciprocating pump, will the modification result in one or more of the following: •A change in the density of the liquid by more than ± 5% from previous maximum/minimum operational experience? •A change to the bulk modulus of the liquid by more than ± 5% from previous maximum/minimum operational experience?	•Pulsation – reciprocating pump (liquid systems only) refer to Section T2.4

There are no reciprocating pumps in the system and therefore no potential issues are identified.

Item 4: Change to operational configuration of positive displacement compressor or pump

Item	Description	If Yes - Potential Issues
4	Will the modification result in a change to the operational configuration of a positive displacement compressor or pump which is outside existing operational experience e.g.: •The use of a second compressor/pump in tandem? •The use of compressor/pump recycle or partial unloading of the compressor?	•Pulsation – reciprocating compressor or pump (liquid and gas systems only) refer to Section T2.4

There are no reciprocating/positive displacement compressors or pumps in the system and therefore no potential issues are identified.

Item 5: Change to centrifugal compressor operational configuration

Item	Description	If Yes - Potential Issues
5	Will the modification result in a centrifugal compressor being operated at low flow conditions?	•Pulsation - rotating stall (gas systems only) refer to Section T2.5

There are no centrifugal compressors in the system and therefore no potential issues are identified.

Item 6: Choked flow and/or sonic velocities

Item	Description	If Yes - Potential Issues
6	Will the modification result in choked flow and/or sonic velocities in the pipework?	•High frequency acoustic excitation (gas systems only) refer to Section T2.7

Choked flow and/or sonic velocities will not occur and therefore no potential issues are identified.

Item 7: Flashing or cavitation

Item	Description	If Yes - Potential Issues
7	Will the modification result in flashing or cavitation?	•Cavitation and Flashing refer to Section T2.9

No issues are anticipated. If the modification had resulted in an increased pressure drop in the system or an increase in liquid temperature then flashing or cavitation could become an issue – however, that is not the case in this example.

Item 8: Change or addition to existing pipework or associated equipment

Item	Description	If Yes - Potential Issues
8	Will the modification result in a change or addition to the existing pipework or associated equipment (valves, machinery or intrusive elements such as thermowells) which is not a like-for-like replacement?	For changes to valves (including change of valve type or changes to valve closing timings) check for: •Surge/Momentum Change refer to Section T2.8 For changes to machinery check for: •Mechanical excitation refer to Section T2.3 For changes to thermowells check for: •Vortex shedding from intrusive elements refer to TM-04 For changes to pipework, supports, small bore connections and tubing check for: •Poor geometry refer to TM-05 and TM-06

No changes are being made to the existing pipework or associated equipment and therefore no potential issues are identified.

D.4 **EXAMPLE D4: SEPARATION SYSTEM: QUANTITATIVE ASSESSMENT**

This example follows on directly from the qualitative assessment undertaken in **Example D3**. From the results obtained in the qualitative assessment the following excitation mechanisms (based on those identified) should be considered for a quantitative assessment using the relevant methods given in Technical Module **TM-02**:

Excitation Mechanism	Technical Module	Section
Flow induced turbulence	TM-02	T2.2
Surge/momentum changes	TM-02	T2.8

Each of these will be addressed in turn.

D.4.1 **Flow Induced Turbulence (see T2.2)**

Step 1: Determine ρv^2 (see T2.2.3.1)

Stream 1 is multiphase, therefore:

$$\rho v^2 = (\text{effective density}) \times (\text{effective velocity})^2$$

Effective density = total mass flow rate / total volumetric flowrate
 = 855976 / (5436+326+532)
 = 136 kg/m³

Pipe diameter (20") = 508mm

Wall thickness (STD) = 9.525mm

Effective velocity = total volumetric flow rate / pipe internal area
 = ((5436+326+532)/3600) / 0.188
 = 9.3m/s

Stream 4 is single phase.

Pipe diameter (10") = 273mm

Wall thickness (STD) = 9.271mm

Velocity = (532/3600)/0.051
 = 2.9 m/s

The values of ρv^2 for streams 1 and 4 are summarised below:

Stream	1	4
Velocity (m/s)	9.3	2.9
Density (kg/m ³)	136	988
Calculated ρv^2 (kg/m.s ²)	11763	8309

Step 2: Determine Fluid Viscosity Factor (see T2.2.3.2)

Streams 1 and 4 are multiphase and liquid respectively and therefore the fluid viscosity factor (FVF) = 1.

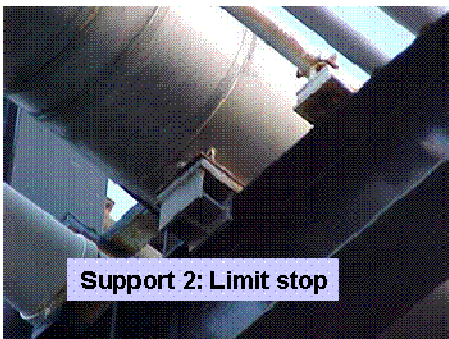
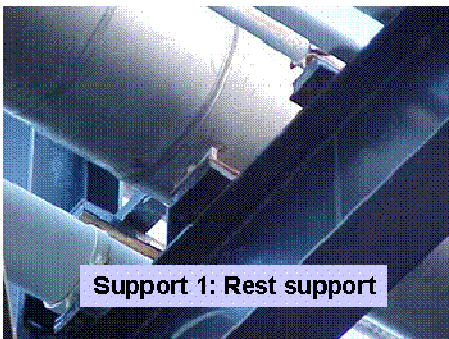
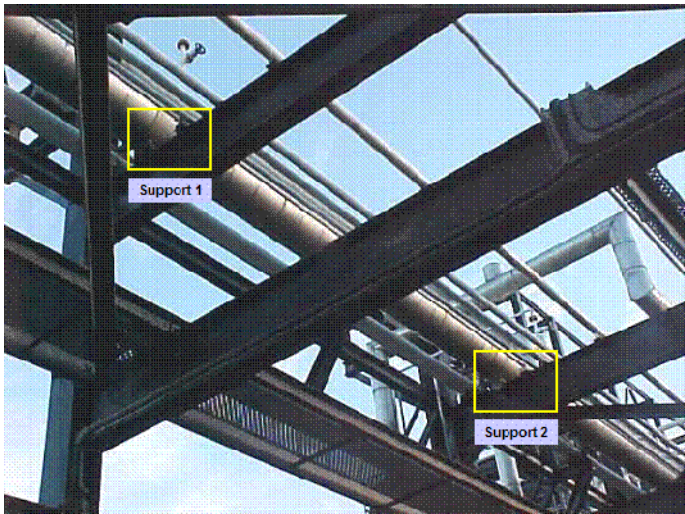
Step 3: Determine Support Arrangement (see T2.2.3.3)

The pipe support arrangement must now be determined. This requires the maximum span lengths between supports to be identified (see guidance in **Appendix B**) and compared with the criteria given in **Table T2-1**. This can be done by working through system isometrics, or as in this case, walking the lines. Alternatively the fundamental natural frequency could be calculated (refer to **Section B.1**) or measuring using modal testing techniques (refer to **Section T8.3**).

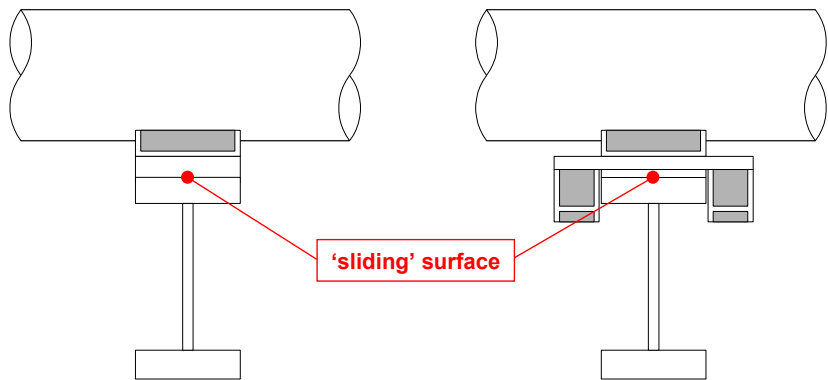
The following assessment uses the method given in **Appendix B**.

Separator Inlet

The separator inlet is a 20" NB pipe (actual outside diameter 508mm), supported at regular intervals on a pipe rack. At the left end there is a rest support (**Support 1**), which is designed to support the pipe vertically. The support at the right end (**Support 2**) is a limit stop that will allow the pipe to move from side to side. In both cases the support is attached to a substantial 'I' girder, which forms part of the pipe rack.

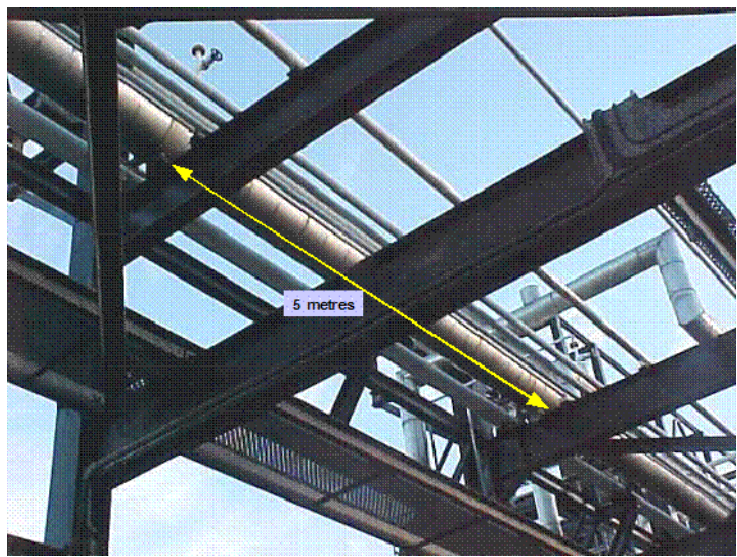


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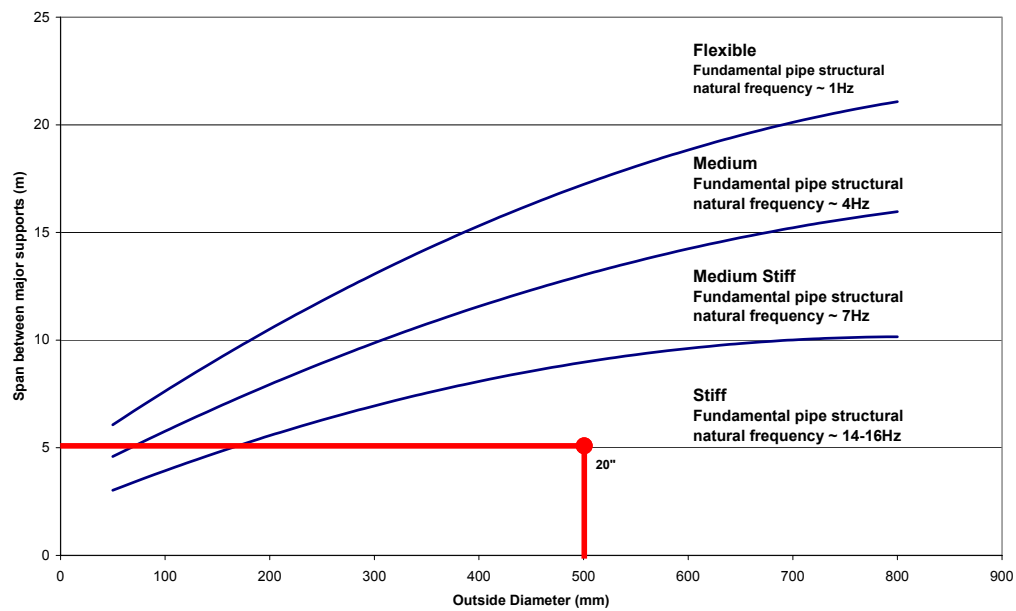
Due to the nature of this type of support, and the self-weight of the pipe, there will be a significant amount of friction between the sliding surfaces. Whilst this friction will not constrain the pipe when subjected to high static loads (e.g. thermal growth), it is usually the case that the friction is sufficient to restrain the pipe when the pipe vibrates.

Both the supports can be considered 'effective', and both have a substantial 'I' girder which forms the primary foundation. The span length can therefore be taken as the distance between these two supports: in this case, approximately 5 metres.



This span length can then be used to determine the support classification as shown below.

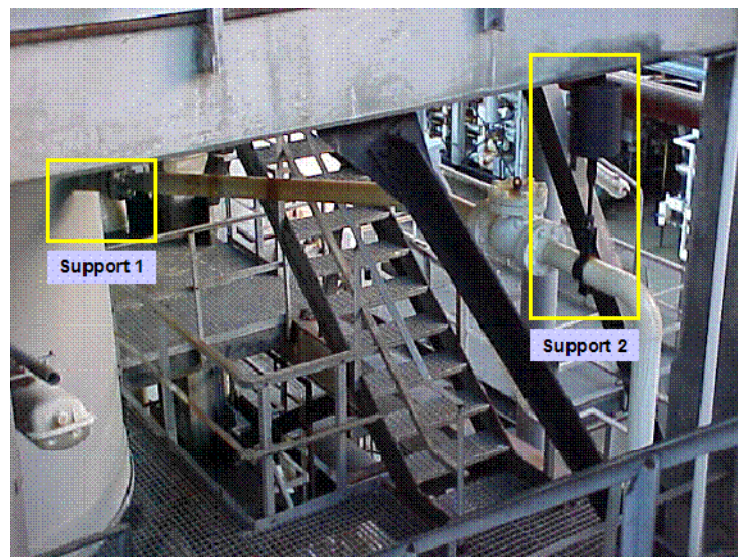
APPENDIX D – WORKED EXAMPLES



This results in a 'stiff support arrangement' classification.

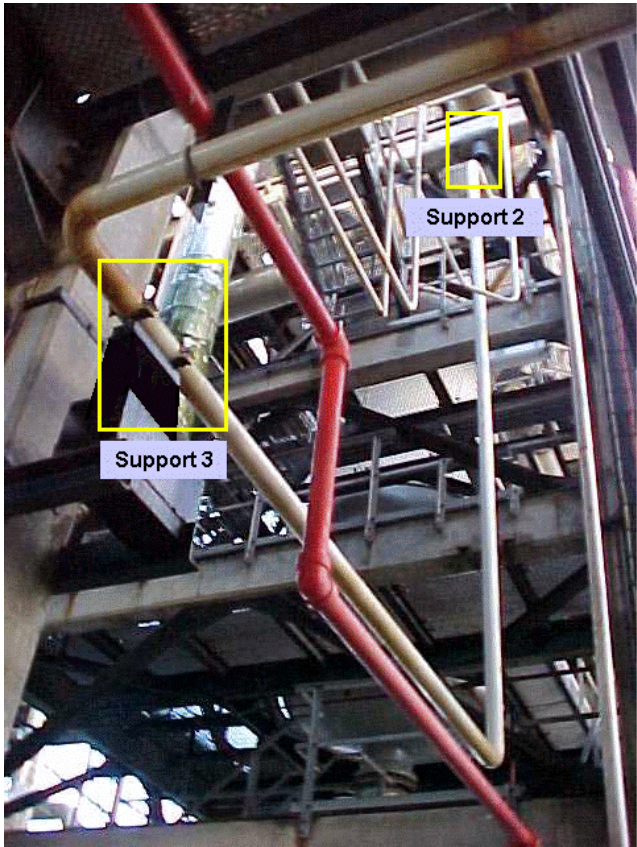
Produced Water Outlet

The separator inlet is a 10” NB pipe (actual outside diameter 273mm); the longest span is immediately downstream of the vessel as shown below.

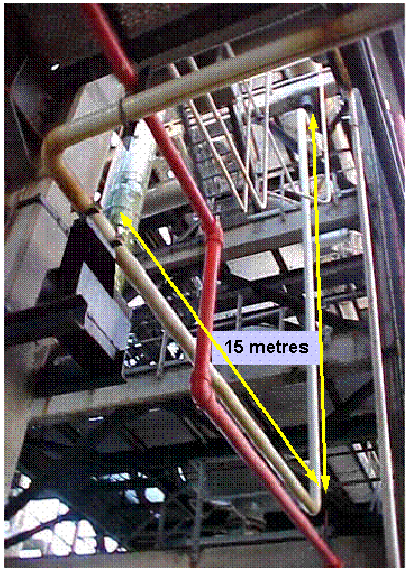
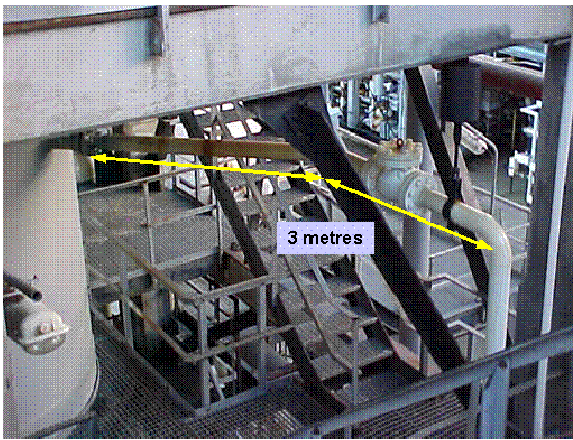


The first support (**Support 1**) is the vessel nozzle, and constitutes a stiff termination point for the pipe. The next support (**Support 2**) is a variable spring hanger with an extended support rod between the spring and the pipe and is therefore not considered an ‘effective’ support..

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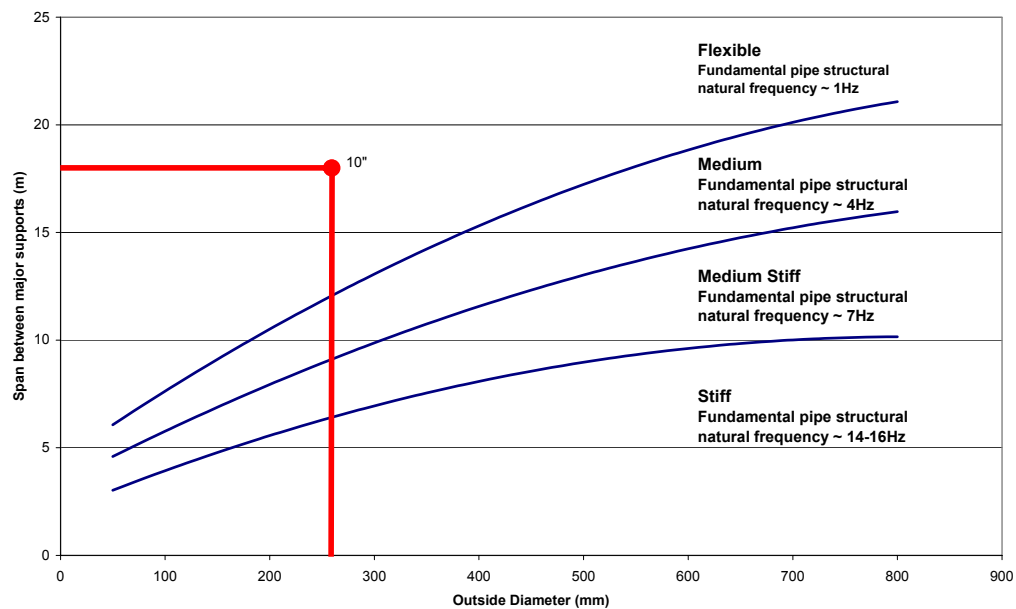


The final support (**Support 3**) is a saddle which itself is well supported and can therefore be considered an ‘effective’ support. The span length is therefore the length between **Support 1** and **Support 3**. This gives a total span length of 18 metres.



This span length can then be used to determine the support classification:

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This results in a “Flexible” classification.

Step 4: Determine Flow Induced Vibration Factor Fv (see T2.2.3.4)

Fv is determined from the expressions given in **Table T2-2** for the relevant pipe outside diameter and support arrangement. The results are summarised below.

Pipe diameter (schedule)	20” (sch STD)	10” (sch STD)
α	894571	60647
β	-0.75085	-0.92703
F_v	45175	2636

Step 5: Calculation of Likelihood of Failure (LOF) (see T2.2.3.5)

Finally the LOF for each line is calculated using:

$$\text{Flow Induced Turbulence L.O.F.} = \frac{\rho v^2}{F_v} FVF$$

The results are summarised below.

Stream (Sub system) Pipe dimensions	1 (supply to separator) 20” (sch STD)	4 (produced water) 10” (sch STD)
ρv^2 (kg/m.s ²)	11763	8309
FVF	1	1
F_v	45175	2636
LOF	0.26	3.15

D.4.2 Surge / Momentum Changes (see T2.8)

Surge and momentum changes only apply when there is an automatically actuated valve. In this example the only valve that meets this criterion on process streams 1 and 4 is FCV1002 on the produced water discharge line from the separator.

[Note: although the oil system line from the separator also has a flow control valve (FCV1001), this particular line has not been identified from the previous qualitative assessment, as there is no change to the oil flow rate].

The assessment should cover liquid or multiphase valve closing (i.e. when the flow control valve shuts in and has the potential to generate a pressure surge transient). Liquid or multiphase valve opening does not apply as the FCV is not a fast acting valve (such as a relief valve).

The procedure shown in **Flowchart T2-7** should be followed:

Step 1: Determine surge pressure (P_{max}) and maximum force (F_{max})

$$P_{max} = \rho c v \quad \text{where } c = \frac{1}{\sqrt{\rho \left(\frac{1}{K} + \frac{D_{ext}}{1000 T E_{ml}} \right)}}$$

In this case:

ρ (fluid density)	= 1000 kg/m ³
K (fluid bulk modulus)	= 2.19 x 10 ⁹ N/m ²
D_{ext}	= 273mm
T (main line wall thickness)	= 9.271mm
E_{ml} (Youngs Modulus of pipe material)	= 207 x 10 ⁹ N/m ²
Therefore c	= 1293 m/s.

The maximum fluid velocity (v) was calculated previously as 2.9 m/s (**Section D4.1**).

Therefore $P_{max} = 1000 \times 1293 \times 2.9 = 3749589$ (N/m²).

$$F_{max} = \rho c v \pi \frac{D_{int}^2}{4 \times 10^9} = 190.7 \text{ kN}$$

The upstream length between the valve and the separator (the first large volume) is approximately 2 metres. As $F_{max} > 1 \text{ kN}$ then the next step is to take into account the valve closure time.

Step 2: Effect of valve closure time

The surge pressure is calculated as follows

$$P_{surge} = P_1 \left(\frac{\Omega^2}{2} + \Omega^2 \sqrt{\frac{1}{4} + \frac{1}{\Omega^2}} \right) \quad \text{where} \quad \Omega = \frac{\rho v L_{up} \phi}{P_1}$$

P_1 (static pressure)	= 5.5 barg = 6.5 x 10 ⁵ N/m ²
v (fluid velocity)	= 2.9 m/s

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L_{up} (upstream line length)

ρ (fluid density)

Valve closure time

The valve is a globe valve, and therefore $\phi = -2.266/T_{close} - 0.32 = -0.7732$

= 2m

= 1000 kg/m³

= 5 seconds

Therefore $\Omega = -0.0069$

and $P_{surge} = 54084 \text{ N/m}^2$

Which in turn gives an F_{max} of 2.75 kN

Step 3: Limit Force Calculation

The next step is to calculate the limit force using:

$F_{lim} = (16.8 \times \Psi^3 - 1.81 \times \Psi^2 + 525 \times \Psi + 25.3) \times D_{ext} \times \theta \times \pi \times D_{int}^2 / (4 \times 10^9) \text{ (kN)}$

Line	10" sch STD
T (mm)	9.271
T sch 40 (mm)	9.271
Ψ	1
D_{ext} (mm)	273
D_{int} (mm)	254.5
θ (flexible support)	0.5
F_{lim} (kN)	3.92

Step 4: LOF Calculation

Finally, the LOF value is calculated using F_{max} / F_{lim}

Line	10" sch STD
F_{max} (kN)	2.75
F_{lim} (kN)	3.92
LOF	0.70

D.4.3 Summary and Interpretation of Main Line LOF Scores

Stream (Sub system) Pipe dimensions	1 (supply to separator) 20" (sch STD)	4 (produced water) 10" (sch STD)
Flow induced turbulence	0.26	3.15
Surge / momentum	n/a	0.70

Main Line LOF ≥ 1.0

A summary of the actions required for a main line LOF score ≥ 1.0 are given below.

Score	Action	Technical Module
LOF ≥ 1.0	The main line shall be redesigned, resupported or a detailed analysis of the main line shall be conducted, and vibration monitoring of the main line shall be undertaken (Note 1)	TM-09 TM-07/TM-08
	Corrective actions shall be examined and applied as necessary	TM-10
	Small bore connections on the main line shall be assessed.	TM-03
	A visual survey shall be undertaken to check for poor construction and/or geometry and/or support for the main line and/or potential vibration transmission to neighbouring pipework.	TM-05 TM-06

- The produced water outlet line scores an LOF of 3.15 for flow induced turbulence due to the combination of a high value of ρv^2 combined with a flexible support arrangement (i.e. a low fundamental structural natural frequency). Changes to the way the pipe is supported – to increase the fundamental structural natural frequency by reducing the long unsupported span – would be one way of reducing the LOF score. This could potentially be achieved by introducing an intermediate support from the lower of the two horizontal deck beams.

In addition, all small bore connections on the line should also be assessed.

Main Line LOF ≥ 0.5 and < 1.0

A summary of the actions required for a main line LOF score ≥ 0.5 are given below.

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Score	Action	Technical Module
1.0 > LOF ≥ 0.5	The main line should be redesigned, resupported or a detailed analysis of the main line should be conducted, or vibration monitoring of the main line should be undertaken (Note 1)	TM-09 TM-07/TM-08
	Corrective actions should be examined and applied as necessary	TM-10
	Small bore connections on the main line shall be assessed.	TM-03
	A visual survey shall be undertaken to check for poor construction and/or geometry and/or support for the main line and/or potential vibration transmission to neighbouring pipework.	TM-05 TM-06

There is one main line that scores an LOF ≥ 0.5:

- The produced water outlet line scores an LOF of 0.70 for pressure surge. In this case any changes to the pipe support arrangement considered for the flow induced turbulence issue (see above) would also be beneficial in terms of reducing the LOF score. A surge analysis (see **Section T9.6**) taking into account the true valve closure characteristics (i.e. valve flow coefficient (C_v) against percentage closure) might also be considered.

Main Line LOF ≥ 0.3 and < 0.5

There are no lines that score an LOF ≥ 0.3 and < 0.5.

Main Line LOF < 0.3

A summary of the actions required for a main line LOF score < 0.3 are given below.

Score	Action	Technical Module
LOF < 0.3	A visual survey should be undertaken to check for poor construction and/or geometry and/or support for the main line and/or potential vibration transmission from other sources.	TM-05 TM-06

For all main lines a walkdown should be conducted to ensure that the as-built arrangement is fit for purpose, and that no changes have been introduced in the period since the original assessment was undertaken, using the guidance given in **TM-06** and **TM-07**.

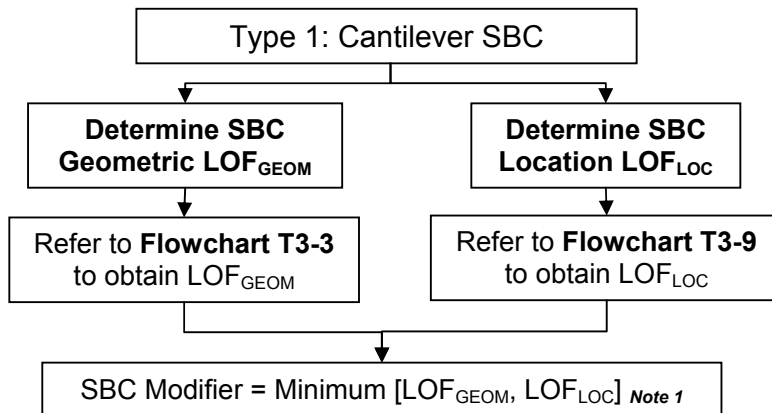
D.5 EXAMPLE D5: SMALL BORE CONNECTION (TYPE 1)

This example is a connection on the compressor discharge line assessed in **Example D2**, which scored a main line LOF of 1.0.



The connection shown is a pressure tapping. There is a single isolation valve, with an instrument line from the flange at the top of the valve. The connection is 1" NB, and the parent pipe schedule is Sch 120. The parent pipe is lagged. The connection is located close to mid span on the parent pipe (i.e. approximately halfway between parent pipe supports).

For a Type 1 SBC **Flowchart T3-2** applies:

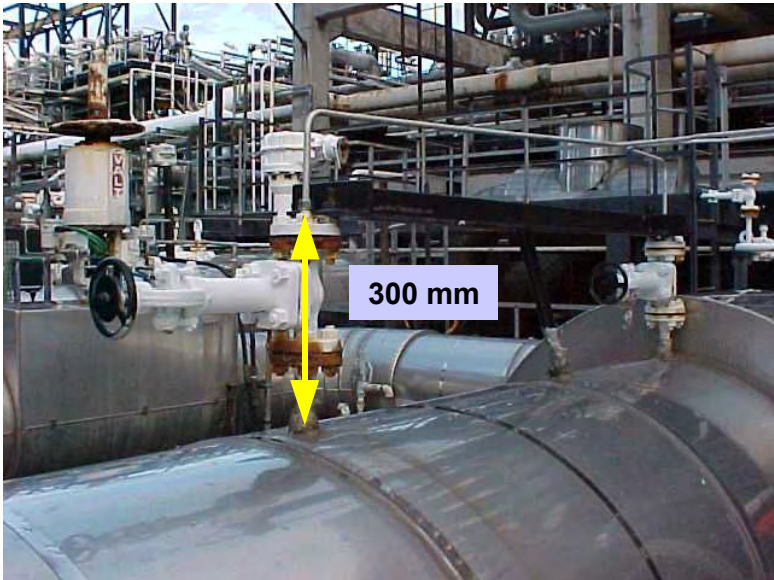


Step 1: Determine LOF_{GEOM} (Flowchart T3-3)

Type of fitting: the lagging also makes it difficult to identify the type of fitting. In this case, the isometric of the parent pipe identified this connection as a weldolet fitting. If no information had been available then the fitting type would have been scored as a 'set on' to provide a conservative assessment.

Overall length of branch: the visible length of the connection is 300 mm. However, an additional length must be included to account for the thickness of the lagging in order to give a 'true' length of the fitting from the wall of the parent pipe to the end of the valve. In this case the lagging is approximately 200 mm deep, so the total length is 500 mm.

Number and size of valves: there is one valve (valve ratings below ANSI 900).



Parent pipe schedule: the parent pipe is Schedule 120. As this is not on the list given in **Flowchart T3-3** then the next lowest 'standard' Schedule is 80 – this is used for the assessment.

SBC minimum diameter: the minimum SBC diameter is 1" NB.

A summary of the scores and the calculated LOF_{GEOM} is given below:

Geometric item	Value	Score
Type of fitting	Weldolet	0.9
Overall length of branch	500mm	0.7
Number & size of valves	1	0.5
Parent pipe schedule	120 (assessed as 80)	0.5
SBC minimum diameter	1" NB	0.7
LOF_{GEOM}		0.66

Step 2: Determine LOF_{LOC} (Flowchart T3-9)

In this case the main line LOF is known and is equal to 1.0. Therefore the LOF_{LOC} defaults to 1.0.

Step 3: Determine SBC Modifier (Flowchart T3-2)

The minimum of the LOF_{GEOM} and LOF_{LOC} scores is taken. In this case this results in a SBC Modifier score of 0.66.

Step 4: Determine SBC LOF (Flowchart 3-4)

The main line LOF is multiplied by 1.42. In this case this results in $1.0 \times 1.42 = 1.42$.

The minimum of this value (1.42) and the SBC Modifier (0.66) is then obtained to give the SBC LOF (0.66).

Score	Action	Technical Module
LOF ≥ 0.7	The SBC shall be redesigned, resupported or a detailed analysis shall be conducted, and vibration monitoring of the SBC shall be undertaken A visual survey shall be undertaken to check for poor construction and/or geometry for the SBC's and instrument tubing.	TM-11 TM-07/TM-08 TM-05/TM-06

Note: in this example the main line assessment (which is detailed in Example D2) has been identified as being associated with tonal excitation from a dead leg branch (the recycle line) and also from mechanical excitation from the compressor. If the excitation frequencies are known then the structural natural frequencies of the SBC should also be determined by specialist measurement or predictive techniques (see **Chapter 3, Section 3.3.3**).

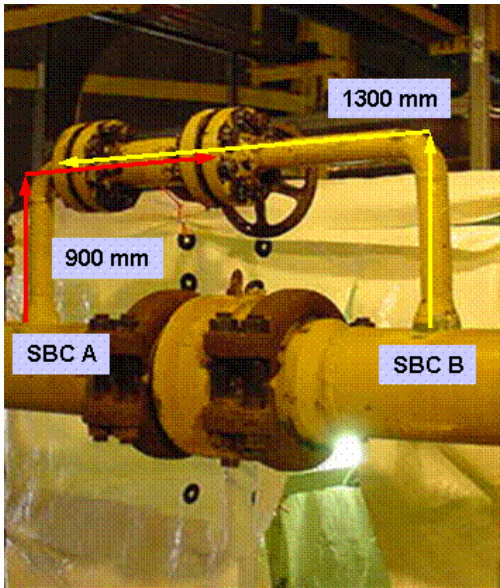
D.6 EXAMPLE D6: SMALL BORE CONNECTION (TYPE 2)

This example is a bypass which exits and enters the same main line.



There is a single valve and the connection is 2" NB, and the parent pipe schedule is Sch 40. The connection is located just downstream of a 90 degree bend in the parent pipe. It is assumed for the case of this example that the mainline LOF has previously been assessed with an LOF of 0.49.

For a Type 2 SBC **Flowchart T3-4** applies. In this case the connection is divided into two (see **C.1.11**) as shown below, each with different total lengths.



Step 1: Determine LOF_{GEOM} (Flowchart T3-4)

Type of fitting: Both connections (A and B) are weldolet fittings.

Overall length of branch: for SBC A the length of the connection is 900 mm. The length of SBC B is 1300mm.

Number and size of valves: for both SBC A and B there is one valve (valve ratings below ANSI 900).

Parent pipe schedule: the parent pipe is Schedule 40.

SBC minimum diameter: the minimum SBC diameter is 2" NB.

Using the method given in **Flowchart T3.3** the following values for LOF_{GEOM(A)} and LOF_{GEOM(B)} are given below:

	SBC A		SBC B	
Geometric item	Value	Score	Value	Score
Type of fitting	Weldolet	0.9	Weldolet	0.9
Overall length of branch	>600mm	0.9	>600mm	0.9
Number & size of valves	1	0.5	1	0.5
Parent pipe schedule	40	0.7	40	0.7
SBC minimum diameter	2" NB	0.5	2" NB	0.5
LOF _{GEOM}		0.7		0.7

The final LOF_{GEOM} applied to the complete connection is the maximum of the LOF_{GEOM} score for SBC A and B which in this case is 0.7.

Step 2: Determine LOF_{LOC} (Flowchart T3-9)

In this case the main line LOF is known and is equal to 0.49. The connection is just downstream of a bend and the parent pipe schedule is 40. This gives values of 0.9 and 0.7 respectively, giving an LOF_{LOC} of 0.8.

Step 3: Determine SBC Modifier (Flowchart T3-4)

The minimum of the LOF_{GEOM} and LOF_{LOC} scores is taken. In this case this results in a SBC Modifier score of 0.7.

Step 4: Determine SBC LOF (Flowchart 3-4)

The main line LOF is multiplied by 1.42. In this case this results in $0.49 \times 1.42 = 0.696$.

The minimum of this value (0.696) and the SBC Modifier (0.7) is then obtained to give the SBC LOF (0.696).

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Score	Action	Technical Module
LOF ≥ 0.7	The SBC shall be redesigned, resupported or a detailed analysis shall be conducted, and vibration monitoring of the SBC shall be undertaken A visual survey shall be undertaken to check for poor construction and/or geometry for the SBC's and instrument tubing.	TM-11 TM-07/TM-08 TM-05/TM-06
0.7 > LOF ≥ 0.4	Vibration monitoring of the SBC should be undertaken. Alternatively the SBC may be redesigned, resupported or a detailed analysis conducted. A visual survey should be undertaken to check for poor construction and/or geometry for the SBC's and instrument tubing.	TM-07/TM-08 TM-11 TM-05/TM-06

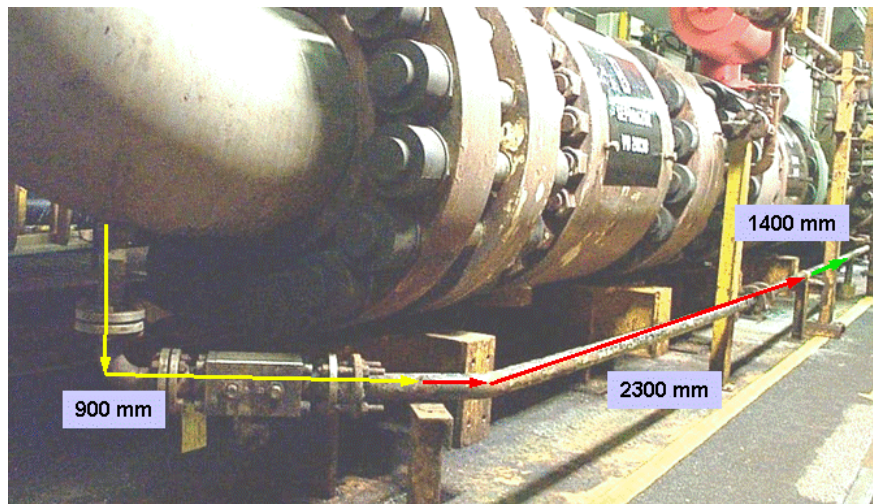
The final result is borderline (i.e. just below 0.7). Consideration should therefore be given to applying some form of modification – in this case bracing back to the main line is a practical option. See **TM-11** for potential options.

D.7 EXAMPLE D7: SMALL BORE CONNECTION (TYPE 3)

This example is a long chemical injection line.



There is a single valve between the parent pipe and the first resting support on the connection. This first resting support is 900mm from the connection to the parent pipe. The span length to the next support on the connection is approximately 2300mm. The connection is 1.5" NB, and the parent pipe schedule is Sch 120. The connection is located just downstream of a 90 degree bend in the parent pipe and close to a fixed anchor on the parent pipe. It is assumed for the case of this example that the mainline LOF is unknown.



For a Type 3 SBC **Flowchart T3-5** applies. In this case the SBC modifier must be obtained for (i) the first span and (ii) the subsequent spans.

First Span

Step 1: Determine LOF_{GEOM} (Flowchart T3-6)

$LOF_{GEOM(C)}$ is obtained as follows:

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As there is a mass associated with the first span $LOF_{GEOM(C)}$ is obtained using **Flowchart T3-3** as follows:

Type of fitting: The connection to the parent pipe is a weldolet fitting.

Overall length of branch: the length of the connection to the end of the valve is approximately 700mm.

Number and size of valves: there is one valve (valve ratings below ANSI 900).

Parent pipe schedule: the parent pipe is Schedule 120.

SBC minimum diameter: the minimum SBC diameter is 1.5" NB.

Using the method given in **Flowchart T3-3** the following value for $LOF_{GEOM(C)}$ is given below:

Geometric item	Value	Score
Type of fitting	Weldolet	0.9
Overall length of branch	700mm	0.9
Number & size of valves	1	0.5
Parent pipe schedule	120 (assessed as 80)	0.5
SBC minimum diameter	1.5" NB	0.6
$LOF_{GEOM(C)}$		0.68

$LOF_{GEOM(D)}$ is obtained as follows:

The first span length (900mm to the first support) is divided by the fitting span factor from **Table 3-1**. For a weldolet fitting the fitting span factor is 0.7; the span length divided by the fitting span factor is therefore $900\text{mm}/0.7 = 1286\text{mm}$.

This value is then assessed against the criteria given in **Figure T3-1**. For the combination of modified span length (1286mm) and connection diameter (1.5") this results in an $LOF_{GEOM(D)}$ of 0.2.

$LOF_{GEOM(E)}$ is obtained as follows:

This time the first span length (900mm to the first support) is multiplied by the fitting span factor from **Table 3-1**. For a weldolet fitting the fitting span factor is 0.7; the span length multiplied by the fitting span factor is therefore $900\text{mm} \times 0.7 = 630\text{mm}$.

The minimum allowable span length from **Table 3-2** is 1.3m for a 1.5" NB connection. This is greater than 630mm and therefore there is a potential issue with respect to the connection being supported from the deck too close to the parent pipe. The $LOF_{GEOM(E)}$ is therefore set to 0.7.

The LOF_{GEOM} is then set to the maximum of $LOF_{GEOM(C)}$, $LOF_{GEOM(D)}$ and $LOF_{GEOM(E)}$, which results in an LOF_{GEOM} of 0.7.

Step 2: Determine LOF_{Loc} (Flowchart T3-9)

In this case the main line LOF is not known and therefore the LOF_{Loc} is set to 1.0.

Step 3: Determine SBC Modifier – first span (Flowchart T3-6)

The minimum of the LOF_{GEOM} and LOF_{LOC} scores is taken. In this case this results in a SBC Modifier score of 0.7.

Subsequent Spans

Step 1: Determine LOF_{GEOM} (Flowchart T3-7)

$LOF_{GEOM(F)}$ is obtained as follows:

The maximum span length associated with a mass is 2300mm. For the combination of span length (2300mm) and connection diameter (1.5") this results in an $LOF_{GEOM(F)}$ of 0.6 from **Figure T3-3**.

$LOF_{GEOM(G)}$ is obtained as follows:

The maximum span length without a mass is 1400mm. For the combination of span length (1400mm) and connection diameter (1.5") this results in an $LOF_{GEOM(G)}$ of 0.2 from **Figure T3-4**.

The LOF_{GEOM} is then set to the maximum of $LOF_{GEOM(F)}$ and $LOF_{GEOM(G)}$, which results in an LOF_{GEOM} of 0.6.

Step 2: Determine LOF_{LOC} (Flowchart T3-7)

LOF_{LOC} is set to 1.0.

Step 3: Determine SBC Modifier – subsequent spans (Flowchart T3-6)

The minimum of the LOF_{GEOM} and LOF_{LOC} scores is taken. In this case this results in a SBC Modifier score of 0.6.

Overall Connection SBC Modifier

Step 1: Determine SBC Modifier (Flowchart T3-5)

The overall SBC Modifier is the maximum of the SBC Modifier [first span] (0.7) and the SBC Modifier [subsequent spans] (0.6). This results in a final overall SBC Modifier for this connection of 0.7.

Step 2: Determine SBC LOF (Flowchart 3-4)

The main line LOF is multiplied by 1.42. In this case this results in $1.0 \times 1.42 = 1.42$.

The minimum of this value (1.42) and the SBC Modifier (0.7) is then obtained to give the SBC LOF (0.7).

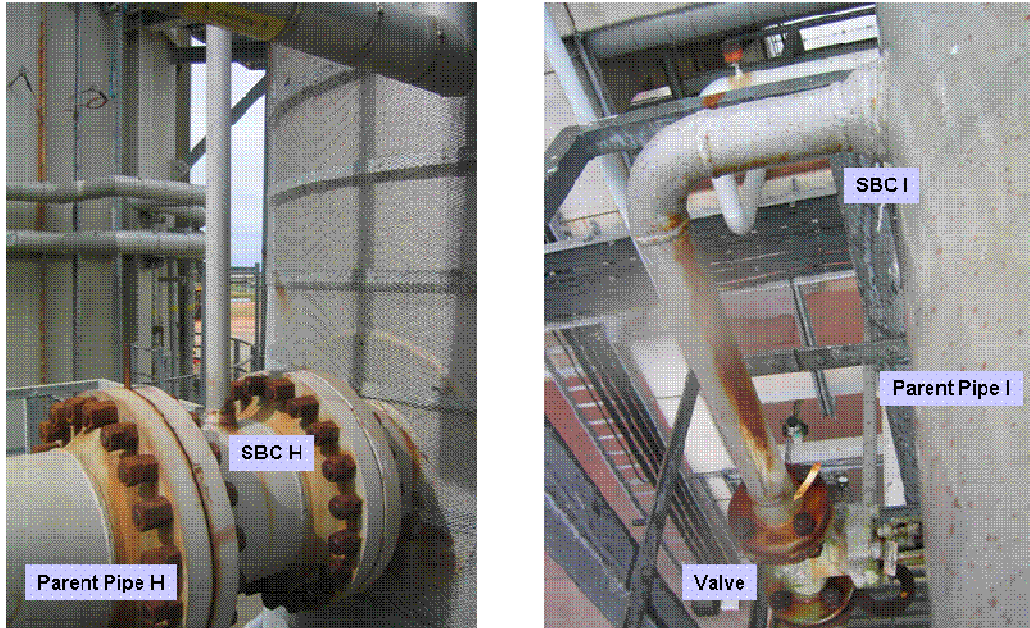
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Score	Action	Technical Module
LOF $\geq$ 0.7	The SBC shall be redesigned, resupported or a detailed analysis shall be conducted, and vibration monitoring of the SBC shall be undertaken A visual survey shall be undertaken to check for poor construction and/or geometry for the SBC's and instrument tubing.	TM-11 TM-07/TM-08 TM-05/TM-06

The final result indicates that some form of modification is required. From the assessment process the main issue which gives rise to the high score is that the first support to the deck is too close to the parent pipe and therefore this support needs to be relocated. Note that the assessment is based on a main line LOF = 1.0 as no main line assessment has been made, and will be conservative.

D.8 EXAMPLE D8: SMALL BORE CONNECTION (TYPE 4)

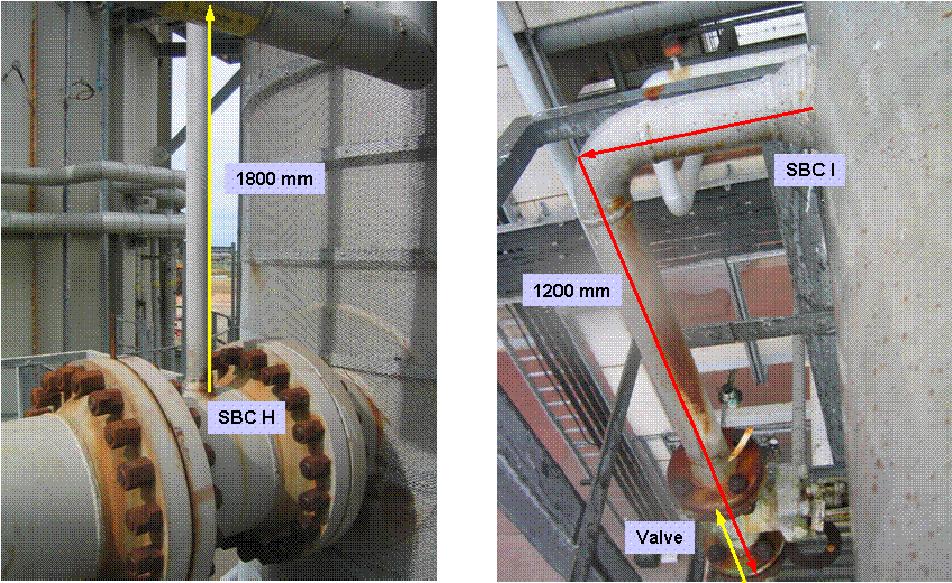
This example is a 2" NB connection which spans between two separate parent pipes. There is no intermediate support on the connection, but there is a valve which isolates one parent main line from the other.



The left hand picture shows the connection from SBC H rising vertically from its parent pipe. The 2" line is then connected to an isolation valve (right hand picture) before turning through 90 degrees and connecting to the other parent pipe at SBC I.

The parent pipe schedule in both cases is Sch 160. SBCs H and I are both located close to fixed supports on their respective parent pipes. It is assumed for the case of this example that the main line LOF has previously been assessed with an LOF of 0.3. The fittings type in both cases is a weldolet.

For a Type 4 SBC **Flowchart T3-8** applies. In this case the connection is divided into two (see **C.1.11**) as shown below, each with different total lengths. The conservative approach is to assume that although the valve is located closer to SBC H than SBC I both SBCs ‘see’ the valve.



Step 1: Determine LOF_{GEOM} (Flowchart T3-8)

LOF_{GEOM(H)} is obtained using **Flowchart T3-3** as follows:

- Type of fitting:** The connection to the parent pipe is a weldolet fitting.
- Overall length of branch:** the length of the connection to the end of the valve is approximately 1800mm.
- Number and size of valves:** there is one valve.
- Parent pipe schedule:** the parent pipe is Schedule 160.
- SBC minimum diameter:** the minimum SBC diameter is 2” NB.

Using the method given in **Flowchart T3.3** the following value for LOF_{GEOM(H)} is given below:

Geometric item	Value	Score
Type of fitting	Weldolet	0.9
Overall length of branch	1800mm	0.9
Number & size of valves	1	0.5
Parent pipe schedule	160	0.3
SBC minimum diameter	2” NB	0.5
LOF _{GEOM(H)}		0.62

LOF_{GEOM(I)} is obtained using **Flowchart T3-3** as follows:

- Type of fitting:** The connection to the parent pipe is a weldolet fitting.
- Overall length of branch:** the length of the connection to the end of the valve is approximately 1200mm.

Number and size of valves: there is one valve (valve ratings below ANSI 900).

Parent pipe schedule: the parent pipe is Schedule 160.

SBC minimum diameter: the minimum SBC diameter is 2" NB.

Using the method given in **Flowchart T3.3** the following value for $LOF_{GEOM(I)}$ is given below:

Geometric item	Value	Score
Type of fitting	Weldolet	0.9
Overall length of branch	1200mm	0.9
Number & size of valves	1	0.5
Parent pipe schedule	160	0.3
SBC minimum diameter	2" NB	0.5
$LOF_{GEOM(I)}$		0.62

$LOF_{GEOM(J)}$ is obtained as follows:

The overall span length (1800 + 1200 mm = 3000mm) is divided by the fitting span factor from **Table 3-1**. For a weldolet fitting the fitting span factor is 0.7; the span length divided by the fitting span factor is therefore 3000mm/0.7 = 4286mm.

This value is then assessed against the criteria given in **Figure T3-1**. For the combination of modified span length (4286mm) and connection diameter (2") this results in an $LOF_{GEOM(J)}$ of 0.6.

$LOF_{GEOM(K)}$ is obtained as follows:

This time the overall span length (3000mm) is multiplied by the fitting span factor from **Table 3-1**. For a weldolet fitting the fitting span factor is 0.7; the span length multiplied by the fitting span factor is therefore 3000mm x 0.7 = 2100mm.

The minimum allowable span length from **Table 3-3** is 2m for a 2" NB connection. This is less than 2100mm and therefore there are no potential issues with respect to the connection being too short to accommodate relative movement between the two parent pipes. The $LOF_{GEOM(K)}$ is therefore set to 0.2.

The LOF_{GEOM} is then set to the maximum of $LOF_{GEOM(H)}$, $LOF_{GEOM(I)}$, $LOF_{GEOM(J)}$ and $LOF_{GEOM(K)}$, which results in an LOF_{GEOM} of 0.62.

Step 2: Determine LOF_{LOC} (Flowchart T3-9)

In this case the main line LOF is 0.3. Both connections are near fixed supports on the parent pipe(s) and the parent pipe schedule is 160. This gives values of 0.1 and 0.3 respectively, giving an LOF_{LOC} of 0.2 for both SBC H and SBC I.

Note: if the two different SBCs had scored different values of LOF_{LOC} then the maximum of the two scores would be taken.

Step 3: Determine SBC Modifier (Flowchart T3-8)

The minimum of the LOF_{GEOM} and LOF_{LOC} scores is taken. In this case this results in a SBC Modifier score of 0.2.

Step 4: Determine SBC LOF (Flowchart 3-4)

The main line LOF is multiplied by 1.42. In this case this results in $0.3 \times 1.42 = 0.426$.

The minimum of this value (0.426) and the SBC Modifier (0.2) is then obtained to give the SBC LOF (0.2).

Score	Action	Technical Module
LOF < 0.4	A visual survey should be undertaken to check for poor construction and/or geometry for the SBC's and instrument tubing.	TM-05/TM-06

A walkdown should be conducted of the SBC to ensure that the as-built arrangement is fit for purpose, using the guidance given in **TM-06** and **TM-07**.

Appendix E

TERMS

Term	Description
API	American Petroleum Institute
Broadband	Energy is input over a wide frequency range
Choked Flow	Choked flow occurs when the gas velocity is the same as the speed of a pressure wave through the fluid and the maximum mass flow rate is achieved.
Dynamic viscosity	Ratio of shear stress to the associated strain rate
Forced vibration	If the frequency of the excitation does not match a natural frequency, then vibration will still be present at the excitation frequency, although at much lower levels than for the resonant case.
HAZID	Hazard Identification
HAZOP	Hazard and Operability
KE	Kinetic Energy
LOF	Likelihood of failure
Mode shape	The relative displacement of all points on a vibrating structure at a given natural frequency.
Natural Frequency	The frequency of free vibration of a system.
Node	A point or line on a vibrating structure that remains stationary.
PWHT	Post weld heat treatment
Ratio of specific heats (C_p/C_v)	The ratio of molar heat capacity at constant pressure to molar heat capacity at constant volume.
Resonance	The resonant frequency of a system is defined as the natural frequency for which the response of the system is a maximum. If the excitation frequency is either increased or decreased the amplitude of response will decrease.
RMS	Root mean squared
SBC	Small Bore Connection
SI	International System of Units
S-N diagram	A plot showing the relationship of stress, S, and the number of cycles, N, before fracture in fatigue testing.

APPENDIX E - TERMS

Term	Description
Stress-intensity factor	A factor, to describe the intensification of applied stress at the tip of a crack.
Tonal	Energy is only input at discrete frequencies
Vapour pressure	The partial pressure of a gas in equilibrium with a condensed form of the same substance.
Vena contractor	The point at which the minimum cross-sectional area of the flow stream occurs